QALA

Universal Solution Management Operating System

Workflows & Use Cases | v2.0 | Universal Edition

Field	Value
Document Type	Workflows and Use Cases — Universal Edition v2.0
System	Qala Universal Solution Management OS
Status	Draft — For Review
Use Case Count	65 use cases across 11 domain chapters
Solution Types	Application, System, Platform, API, Tool, Toolchain, SDK, Library, Package, Good, CPG, Capital Good, Product, Product Line, Service, Managed Service, Consulting Service, Financial Instrument, Investment Solution, Capital Solution, Tax Solution, Agricultural Solution, Business Solution, Process Solution, Research Asset, Resource, Asset, Template, Framework — all solution types
Domains	Software & Technology | Physical Goods & CPG | Financial Solutions | Agricultural & Agribusiness | Professional Services | Business & Operational | Research & Academic | Tools & Libraries | Legal, Tax & Regulatory | Platform & Ecosystem | Cross-Domain Universal Workflows
Companion Docs	SDD v2.0 | PRD v2.0 | RFC v2.0 | Requirements v2.0
Audience	Product, Engineering, QA, Domain Experts, Customers — All Industries

1 Introduction

This document defines the complete set of workflows and use cases for the Qala Universal Solution Management Operating System. Every workflow is presented as a structured use case with actors, trigger, step-by-step procedure, system responses, events published, and success criteria — regardless of solution type or industry.

The 65 use cases in this document demonstrate that Qala's ten capability pillars apply coherently and consistently across every solution type. A pharmaceutical formulation chemist updating an ingredient specification follows the same governed Change Control Request workflow as a software developer updating a dependency. A tax product manager activating a new legislative rule set follows the same effective-date release workflow as a software team deploying to production. The workflows are universal; the domain vocabulary adapts.

> *Reading guide: Each chapter covers a specific solution domain. Within each chapter, use cases are ordered by lifecycle stage: Environment → Configuration → Development → Testing → Release → Distribution → Governance. Cross-domain workflows in Chapter 12 demonstrate how Qala unifies multiple solution types in a single governed process.*

1.1 Workflow Conventions

Field	Description
UC-ID	Unique use case identifier: [Domain Code]-[Sequence]. e.g. UC-SW-001, UC-CPG-002.
Actors	Human roles and system actors participating in the workflow.
Trigger	The event or condition that initiates this workflow.
Outcome	The verifiable state of the system when the workflow succeeds.
Steps	Numbered, actor-attributed actions with system responses.
Events	Platform events published to the event bus at each step.
Alternatives	Notable alternative or exception flows.
Metrics	Key performance indicators for this workflow's success.

1.2 Domain Chapters and Use Case Index

Chapter	Domain	Use Cases	Solution Types Covered
2	Software & Technology	UC-SW-001 to UC-SW-008	Application, System, Platform, API, Microservice, Firmware
3	Physical Goods, CPG & Manufacturing	UC-CPG-001 to UC-CPG-007	Good, CPG, Capital Good, Product, Product Line, Hardware, Component
4	Financial Solutions	UC-FIN-001 to UC-FIN-007	Financial Instrument, Investment Solution, Capital Solution, Insurance Product, Tax Solution

5	Agricultural & Agribusiness	UC-AGRI-001 to UC-AGRI-006	Agricultural Solution, Crop System, Livestock System, Supply Chain Platform, AgTech Platform
6	Professional Services & Consulting	UC-SVC-001 to UC-SVC-006	Service, Managed Service, Professional Service, Consulting Service, Subscription Service
7	Business, Process & Operational	UC-BIZ-001 to UC-BIZ-005	Business Solution, Process Solution, Operational Solution, Workflow Solution
8	Research, Academic & Scientific	UC-RES-001 to UC-RES-006	Research Asset, Dataset, Experimental Framework, Computational Model, Publication Pipeline
9	Tools, Libraries & Packages	UC-TOOL-001 to UC-TOOL-005	Tool, Toolchain, SDK, Library, Package, Framework, Reference Architecture
10	Legal, Tax & Regulatory	UC-LEG-001 to UC-LEG-005	Legal Solution, Regulatory Product, Compliance Framework, Tax Solution
11	Platform & Ecosystem	UC-PLT-001 to UC-PLT-004	Platform, Marketplace, Ecosystem, Multi-Sided Network, Developer Ecosystem
12	Cross-Domain & Universal Workflows	UC-XD-001 to UC-XD-006	All solution types — workflows that span multiple domains

1.3 The Universal Solution Lifecycle

Every use case in this document maps to one or more stages of the Universal Solution Lifecycle. Regardless of solution type, every solution moves through these stages, and Qala governs each transition:

Stage	Description	Universal Activity	All Solution Types
1. Initiate	A new solution or version is proposed and registered.	Solution record created; SDE provisioned; owner assigned.	Yes
2. Configure	The solution and its environment are defined and versioned.	Configuration versioned; environment definition locked.	Yes
3. Develop	The solution is built, formulated, designed, or structured.	Changes governed by CCR; all artefacts versioned in registry.	Yes
4. Test	The solution is validated against its specification.	Test plans executed; quality gates evaluated; defects tracked.	Yes
5. Review	Formal review and approval prior to release.	Approvers sign off; release record assembled; gate checklist complete.	Yes
6. Release	The solution version is released and made available.	Release event published; distribution channels notified.	Yes

7. Distribute	The released solution is delivered to consumers or channels.	Distribution records created; consumer notification sent.	Yes
8. Govern	The live solution is monitored, audited, and maintained.	Drift detected; compliance evidence generated; AI recommendations issued.	Yes
9. Retire	The solution version or the solution itself is decommissioned.	Deprecated state set; consumer migration window opened; archive event.	Yes

2 Software & Technology Solution Workflows

Use cases covering the complete lifecycle of software solutions: applications, systems, platforms, APIs, microservices, and firmware. These are the canonical workflows from which all other domain workflows adapt.

2.1 UC-SW-001: Provision a Software Development SDE

UC-SW-001 — Provision a Software Development SDE

Domain: Software & Technology **Actors:** Developer, Tech Lead, Platform Admin

Trigger: Developer requests a new SDE for an application or service.

Outcome: A hermetic, security-hardened SDE is active; developer can commit, build, and test within 10 minutes.

Step	Actor	Action	System Response	Event Published
1	Developer	Submits SDE Request via portal: solution type = Application, language stack, SDE template, resource tier.	Platform validates request. Checks resource quota. Routes to Tech Lead for approval if required by Factory policy.	SDE_REQUEST_SUBMITTED
2	Tech Lead	Reviews request. Approves in portal.	Platform triggers SDE provisioning from the approved template in the Solution Factory catalogue.	SDE_PROVISIONING_STARTED
3	Platform	—	OCI container image built from locked Nix/Dockerfile definition. Dependencies resolved from approved registries with digest pinning.	BUILD_STARTED → BUILD_PASSED
4	Platform	—	Security baseline policy applied: no ambient credentials, no raw internet egress, SAST scanner injected, SEM telemetry agent registered.	SECURITY_BASELINE_APPLIED
5	Platform	—	SDE activated. Developer notified with access credentials and IDE integration instructions.	SDE_CREATED
6	Developer	Connects IDE to SDE. Runs smoke test build.	Platform records first successful build event. SDE health confirmed active.	BUILD_PASSED

Metric	Target
SDE provisioning time	< 10 minutes from approval
Hermetic build success on first run	> 99%

Security baseline compliance	100% on provision

2.2 UC-SW-002: Hermetic CI/CD Pipeline Execution

UC-SW-002 — Hermetic CI/CD Pipeline Execution

Domain: Software & Technology **Actors:** Developer, CI/CD Engine, SEM, Test Management Service

Trigger: Developer pushes a code commit to the main branch.

Outcome: All pipeline stages pass; SLSA attestation generated; artefacts stored in registry with integrity digests; quality gate result published.

Step	Actor	Action	System Response	Event Published
1	Developer	Commits and pushes code to main branch.	Pipeline triggered automatically by commit webhook. Source commit SHA recorded.	BUILD_STARTED
2	CI/CD Engine	—	Hermetic build container provisioned with digest-pinned dependencies. Source fetched via authenticated clone.	—
3	CI/CD Engine	—	Dependency audit: SCA scans all dependencies against CVE database. Any Critical or High finding fails the build.	DEPENDENCY_AUDIT_PASSED / FAILED
4	CI/CD Engine	—	Code compilation and unit test execution. Test results published to Test Management Service.	TEST_SUITE_COMPLETED
5	CI/CD Engine	—	SAST scan executed. Findings categorised by severity. Critical findings fail stage.	SAST_COMPLETED
6	CI/CD Engine	—	Integration tests executed in isolated test environment provisioned from locked definition.	INTEGRATION_TEST_PASSED
7	CI/CD Engine	—	SLSA provenance attestation generated and signed: source digest, env digest, dependency manifest, build tool version, signer identity.	ATTESTATION_CREATED
8	CI/CD Engine	—	Container image and artefacts stored in Artefact Registry with SHA-256 digests.	ARTEFACT_PUBLISHED
9	Platform	—	Quality gate evaluated: coverage threshold, SAST pass, no critical bugs,	QUALITY_GATE_PASSED

			attestation valid. Gate result published.	
10	Developer	Reviews pipeline results in portal.	Developer notified of pass/fail via notification service.	BUILD_EVENTS_SUMMARY

Alternative flow — Pipeline failure: Any stage failure halts pipeline. Developer notified with failing stage, output log link, and AI-generated root cause suggestion. Defect auto-created for any new SAST finding above threshold.

2.3 UC-SW-003: Governed Software Release with Blue/Green Deployment

UC-SW-003 — Governed Software Release with Blue/Green Deployment

Domain: Software & Technology **Actors:** Release Manager, Tech Lead, QA Engineer, Developer, Approvers

Trigger: Tech Lead declares a release candidate and initiates the release process.

Outcome: Software version is live in production via Blue/Green deployment. Previous version retained as instant rollback target.

Step	Actor	Action	System Response	Event Published
1	Tech Lead	Creates Release Record in portal: version, linked artefact digests, linked CCRs resolved, linked defects fixed.	Platform pre-validates: confirms all linked artefacts have valid SLSA attestations.	RELEASE_PLANNED
2	Platform	—	Release gate checklist evaluated automatically: Test Coverage Gate, SAST Gate, No Critical Defects Gate, Performance Gate, Attestation Gate.	GATE_EVALUATION_STARTED
3	QA Engineer	Reviews gate results. Signs off QA approval in portal.	Approval recorded with timestamp and identity. Compliance evidence record created.	APPROVAL_RECORDED
4	Release Manager	Reviews full release record. Signs off release approval.	All required approvals collected. Release state → Approved.	RELEASE_APPROVED
5	Platform	—	Green environment provisioned as mirror of Blue. New version deployed to Green.	DEPLOYMENT_STARTED
6	Platform	—	Automated smoke tests and synthetic monitors run against Green. Health metrics validated.	DEPLOYMENT_HEALTH_CHECKED

7	Release Manager	Initiates traffic switch in portal.	Load balancer switches 100% traffic from Blue to Green. Blue retained for rollback.	DEPLOYED
8	Platform	—	Post-deployment monitoring active: error rate, latency, business metrics watched for 30 minutes. AI anomaly detection running.	MONITORING_ACTIVE
9	Release Manager	Confirms release stable after monitoring window.	Blue environment scheduled for archive. Distribution record created. Release state → Live.	DISTRIBUTION_EVENTS_PUBLISHED

Rollback flow: If monitoring detects breach of error rate or latency threshold, automatic rollback triggered: traffic returned to Blue within 60 seconds. Release Manager notified. Rollback CCR auto-created for audit trail.

2.4 UC-SW-004: Security Incident — SDE Quarantine and Forensic Recovery

UC-SW-004 — Security Incident — SDE Quarantine and Forensic Recovery

Domain: Software & Technology **Actors:** SEM, Security Engineer, Platform Admin, Developer

Trigger: SEM detects anomalous behaviour in an active SDE (e.g. unexpected network egress, process injection, credential access pattern).

Outcome: Threat contained. Forensic snapshot preserved. Clean SDE re-provisioned from last known-good snapshot. Incident fully documented.

Step	Actor	Action	System Response	Event Published
1	SEM	—	Behavioural anomaly detected. Threat score exceeds quarantine threshold.	THREAT_DETECTED
2	SEM	—	SDE automatically quarantined within 90 seconds: network egress severed, write access suspended, forensic snapshot captured.	SDE_QUARANTINED
3	Platform	—	Security Engineer and SDE owner notified immediately via all registered channels.	SECURITY_ALERT_SENT
4	Security Engineer	Reviews forensic snapshot in secure viewer. Identifies threat vector.	Platform records investigation actions in audit trail. Threat analysis stored as artefact.	INVESTIGATION_STARTED

5	Security Engineer	Confirms threat. Initiates clean SDE recovery.	Platform provisions new SDE from last valid pre-quarantine snapshot. Snapshot integrity verified by digest check.	SDE_ROLLBACK_INITIATED
6	Platform	—	New SDE passes security baseline scan. Developer access restored to clean environment.	SDE_CREATED (clean replacement)
7	Security Engineer	Documents incident findings. Submits post-incident review.	Post-incident review recorded. Security policy updated if systemic gap identified (triggers policy change CCR).	INCIDENT_CLOSED

2.5 UC-SW-005: API Breaking Change Governance

UC-SW-005 — API Breaking Change — Change Control and Consumer Impact

Domain: Software & Technology **Actors:** Developer, Tech Lead, API Consumers, Release Manager

Trigger: Developer proposes a breaking change to a public API.

Outcome: Breaking change implemented in governed way: all consumers identified and notified; deprecation window set; old and new versions served simultaneously during migration window.

Step	Actor	Action	System Response	Event Published
1	Developer	Raises CCR in portal: type = API Breaking Change. Attaches change specification and migration guide.	Platform computes risk score: Breaking Change = High risk. Routes to Tech Lead + Release Manager for approval.	CCR_SUBMITTED
2	Platform	—	Consumer impact analysis: platform queries solution relationship graph for all solutions consuming this API. Impact list generated.	IMPACT_ANALYSIS_COMPLETE
3	Tech Lead	Reviews impact list. Sets deprecation window (minimum 90 days). Approves CCR.	CCR state → Approved. Deprecation schedule recorded.	CCR_APPROVED
4	Developer	Implements breaking change in new API version alongside existing version (parallel versioning).	Both API versions tracked in Artefact Registry. Version routing rules updated.	CHANGE_IMPLEMENTED
5	Platform	—	Consumer notification pipeline: all identified consumers receive notification with change summary, migration guide, and deprecation deadline.	CONSUMER_NOTIFICATIONS_SENT

6	Platform	—	Old API version marked as Deprecated in solution registry. Deprecation date prominently displayed in portal and API response headers.	SOLUTION_DEPRECATED
7	Release Manager	At deprecation deadline: reviews consumer migration status. Confirms >95% consumers migrated. Approves old version retirement.	Old API version state → Retired. Removed from routing.	SOLUTION_RETIRED

2.6 UC-SW-006: Canary Release for a Platform Service

UC-SW-006 — Canary Release — Progressive Traffic Shift with AI Monitoring

Domain: Software & Technology **Actors:** Release Manager, Platform (automated), AI Agent

Trigger: A high-risk platform service update is released with a canary strategy.

Outcome: New version served to 100% of traffic after progressive validation; no degradation in any monitored metric.

Step	Actor	Action	System Response	Event Published
1	Release Manager	Initiates release with strategy = Canary. Configures: initial canary = 5%, increment = 10%, increment interval = 10 minutes, rollback threshold = error rate > 0.5%.	Platform activates canary routing. 5% traffic sent to new version.	DEPLOYED (canary_pct=5)
2	AI Agent	—	Real-time analysis of canary metrics: error rate, latency p99, business metric (conversion, transaction rate). Compares canary vs. baseline.	AI_MONITORING_ACTIVE
3	Platform	—	After 10 minutes with metrics within threshold: canary auto-incremented to 15%.	DEPLOYMENT_INCREMENTED
4	AI Agent	—	Detects latency increase at p99 on specific endpoint during 35% canary phase. Raises alert.	ANOMALY_DETECTED
5	Release Manager	Reviews AI alert. Determines acceptable. Acknowledges and continues.	Acknowledgement recorded. Canary continues.	ANOMALY_ACKNOWLEDGED

6	Platform	—	Canary progresses through 50%→75%→100% over subsequent intervals. All metrics within threshold throughout.	DEPLOYMENT_COMPLETE
7	Release Manager	Confirms release stable. Previous version archived.	Distribution record updated. Release state → Live.	DISTRIBUTION_EVENTS_PUBLISHED

2.7 UC-SW-007: Dependency Vulnerability Response

UC-SW-007 — Dependency Vulnerability Response — CVE Detection to Patched Deploy

Domain: Software & Technology **Actors:** SEM, Security Engineer, Developer, Tech Lead

Trigger: SEM detects a newly published CVE affecting a dependency used in one or more active SDEs.

Outcome: All affected SDEs patched with updated dependency. Patched artefacts re-attested and redistributed within SLA.

Step	Actor	Action	System Response	Event Published
1	SEM	—	New CVE published to NVD. SEM cross-references against all SDE dependency manifests in the Artefact Registry.	VULNERABILITY_FOUND
2	SEM	—	All affected SDEs and solutions identified. CVE severity assessed (CVSS score). Critical: SLA = 24h; High: 72h; Medium: 30 days.	VULNERABILITY_IMPACT_ASSESSED
3	Platform	—	Security Engineer and affected SDE owners notified with: CVE ID, severity, affected package, patched version, recommended action.	SECURITY_ALERT_SENT
4	Developer	Reviews notification. Updates dependency to patched version in SDE. Runs full test suite.	Build triggered with updated dependency. SCA confirms CVE resolved.	BUILD_STARTED → BUILD_PASSED
5	CI/CD Engine	—	SLSA attestation regenerated with patched dependency digests.	ATTESTATION_CREATED
6	Tech Lead	Reviews security fix. Fast-tracks through abbreviated release process (security patch track).	Release record created with security patch type. Gates evaluated: SAST pass, attestation valid, security sign-off.	RELEASE_APPROVED

7	Platform	—	Patched artefact deployed. Distribution record updated. CVE status on all affected solutions → Resolved.	DEPLOYED → VULNERABILITY_RESOLVED

2.8 UC-SW-008: AI-Assisted Test Generation and Quality Improvement

UC-SW-008 — AI-Assisted Test Generation

Domain: Software & Technology **Actors:** AI Agent, Developer, QA Engineer

Trigger: Developer completes a significant feature implementation and commits to development branch.

Outcome: AI-generated test cases covering the new feature and regression risk areas accepted into the test suite; quality gate coverage threshold met.

Step	Actor	Action	System Response	Event Published
1	Developer	Commits feature implementation. Marks branch as ready for test generation in portal.	AI Agent triggered. Analyses diff against previous version. Identifies changed modules, new code paths, and historically bug-prone adjacent areas.	AI_ANALYSIS_STARTED
2	AI Agent	—	Generates candidate test cases: unit tests for new functions, integration tests for changed APIs, regression tests for adjacent risk areas. Test cases annotated with rationale and expected coverage contribution.	AI_RECOMMENDATION_PUBLISHED
3	QA Engineer	Reviews AI-generated test cases in portal. Accepts 18 of 22 candidates. Modifies 3. Rejects 1 with reason.	Accepted and modified tests added to test suite. Rejection feedback stored for AI model tuning.	AI_FEEDBACK_RECORDED
4	CI/CD Engine	—	Test suite executed with new tests included. Coverage report generated.	TEST_SUITE_COMPLETED
5	Platform	—	Quality gate: coverage threshold now met (was 78%, now 83% with new tests). Gate passed.	QUALITY_GATE_PASSED
6	AI Agent	—	Defect prediction model runs against the new feature delta. Flags two functions as elevated risk based on historical patterns from similar code changes across all platform solutions.	PREDICTION_ISSUED

7	Developer	Reviews defect predictions. Decides to add additional defensive assertions to the two flagged functions.	Additional commits made. Pipeline re-runs and passes.	BUILD_PASSED

3 Physical Goods, CPG & Manufacturing Workflows

Use cases covering the lifecycle of physical solutions: consumer packaged goods, pharmaceutical formulations, manufactured products, capital equipment, product lines, and assemblies. The Qala Solution Model governs the information layer; physical production systems integrate via open API.

3.1 UC-CPG-001: New Formulation Development and Regulatory Approval

UC-CPG-001 — New Formulation Development and Regulatory Approval

Domain: Physical Goods & CPG **Actors:** Formulation Chemist, QA Manager, Regulatory Affairs Manager, Legal

Trigger: R&D team initiates development of a new consumer product formulation.

Outcome: Formulation specification v1.0 approved, all regulatory pre-checks passed, artefact stored with attestation, ready for manufacturing specification.

Step	Actor	Action	System Response	Event Published
1	Formulation Chemist	Creates new Solution record in portal: type = Consumer Packaged Good. Enters initial formulation parameters in Lab SDE.	Platform creates solution record with unique ID. Lab SDE provisioned with formulation simulation tools. Initial specification version 0.1 created.	SOLUTION_CREATED → SDE_CREATED
2	Formulation Chemist	Iterates formulation in simulation environment. Each parameter change raises a CCR with type = Formulation Parameter Change.	Platform records each CCR: changed ingredient, changed quantity, rationale. All changes immutably tracked.	CCR_SUBMITTED → CCR_APPROVED (per iteration)
3	Platform	—	Automated regulatory pre-checks run on current formulation: EU REACH substance restrictions, FDA prohibited ingredient list, IFRA compliance, local market rules.	COMPLIANCE_CHECK_PASSED
4	QA Manager	Reviews formulation against internal quality specification. Creates QA test plan: stability study, efficacy test, safety assessment.	Test plan stored in Test Management Service linked to formulation version.	TEST_PLAN_CREATED
5	QA Manager	Records test results from physical laboratory (imported via Quality Event Ingestion API from LIMS).	Platform updates quality record for formulation. Stability test results stored as artefact.	QUALITY_EVENTS_INGESTED

6	Regulatory Affairs Manager	Reviews regulatory pre-check results and lab test results. Prepares regulatory submission package.	Platform generates regulatory submission artefact compiling: formulation specification, ingredient provenance, test results, compliance evidence.	ARTEFACT_PUBLISHED
7	Regulatory Affairs Manager	Submits to regulatory authority (e.g. FDA/EFSA). Records submission reference in platform.	Regulatory gate status = Submitted. Awaiting approval.	REGULATORY_GATE_UPDATED
8	Regulatory Affairs Manager	Records regulatory approval received. Attaches approval document.	Regulatory gate = Approved. Formulation lifecycle state → Approved. Ready for Manufacturing Specification stage.	SOLUTION_STATE_CHANGED

Alternative — Regulatory rejection: Rejection reason recorded as artefact. New CCR raised to address the rejection finding. Formulation revision cycle restarts.

3.2 UC-CPG-002: Ingredient Substitution Change Control

UC-CPG-002 — Ingredient Substitution — Change Control and Impact Assessment

Domain: Physical Goods & CPG **Actors:** Formulation Chemist, QA Manager, Regulatory Affairs Manager, Supply Chain Director, CCB

Trigger: A key ingredient becomes unavailable. Supply chain director requests a substitution.

Outcome: Ingredient substitution approved through CCB. All affected products identified. Regulatory re-assessment completed. Manufacturing updated.

Step	Actor	Action	System Response	Event Published
1	Supply Chain Director	Raises CCR: type = Ingredient Substitution. Identifies original ingredient, proposed substitute, reason (supply disruption), proposed effective date.	Platform computes risk score: Ingredient Substitution = High risk. Routes to CCB.	CCR_SUBMITTED
2	Platform	—	Cascade impact analysis: queries solution relationship graph for all products using the affected ingredient. Returns list of 14 affected SKUs across 3 product lines.	IMPACT_ANALYSIS_COMPLETE
3	Formulation Chemist	Reviews substitute properties. Runs simulation of substitute in Lab SDE. Confirms functional equivalence	Simulation results stored as artefact. 3 SKUs flagged for separate reformulation workflow.	ARTEFACT_PUBLISHED

		for 11 of 14 SKUs. 3 SKUs require reformulation.		
4	CCB	Convenes to review CCR, impact analysis, and simulation results. Approves substitution for 11 SKUs. Approves separate reformulation track for 3 SKUs.	CCR state → Approved. Approval recorded with CCB member identities and rationale.	CCR_APPROVED
5	Regulatory Affairs Manager	Assesses whether substitution requires regulatory notification or re-registration per affected market rules.	Regulatory obligation recorded. Notifications prepared for markets requiring them.	COMPLIANCE_EVENTS_PUBLISHED
6	Formulation Chemist	Updates formulation specification for all 11 approved SKUs. Each update creates a new specification version.	11 new formulation versions created in Artefact Registry. Change provenance traceable to this CCR.	ARTEFACT_PUBLISHED (×11)
7	QA Manager	Runs abbreviated stability and compatibility validation on updated formulations.	QA results imported. Quality gate passed for all 11 updated SKUs.	QUALITY_GATE_PASSED

3.3 UC-CPG-003: Batch Release — From Manufacturing to Retail Distribution

UC-CPG-003 — Batch Release — From Specification to Retail Distribution

Domain: Physical Goods & CPG **Actors:** QA Manager, Release Manager, Manufacturing Team, Distribution Manager

Trigger: A production batch completes manufacturing and is submitted for batch release approval.

Outcome: Batch released with full provenance record. Distribution to retail channels initiated. All batch records stored and traceable.

Step	Actor	Action	System Response	Event Published
1	Manufacturing Team	Submits batch record via ERP integration: batch ID, production date, equipment IDs, in-process quality checks, raw material lot numbers.	Batch record imported as artefact. Platform links batch ID to the approved formulation specification version.	ARTEFACT_PUBLISHED (batch_record)
2	QA Manager	Records finished product quality test results: appearance, assay, microbiological,	Quality events ingested. Quality gate evaluated: all parameters within specification.	QUALITY_GATE_PASSED

		physical parameters. Imports CoA from LIMS.		
3	QA Manager	Reviews batch record and quality results. Provides Qualified Person (QP) sign-off.	QP approval recorded with electronic signature and timestamp. Regulatory-grade audit evidence created.	APPROVAL_RECORDED
4	Release Manager	Creates batch release record: links batch ID, specification version, QA approval, distribution plan.	Release record assembled. Distribution gate checks: all regulatory clearances valid, distribution destinations confirmed.	RELEASE_APPROVED
5	Distribution Manager	Initiates distribution to retail channel in portal: assigns batch to distribution centres, records quantities, carrier details.	Distribution events published. GS1 EPCIS events generated for retail partner integration. Distribution record created with full provenance chain: raw material → manufacturing → QA → retail.	DISTRIBUTION_EVENTS_PUBLISHED
6	Platform	—	Batch lifecycle state → Distributed. Batch provenance record complete and immutably stored. Queryable in < 5 minutes for any recall or traceability request.	SOLUTION_STATE_CHANGED

3.4 UC-CPG-004: Product Recall — Batch Identification and Supply Chain Withdrawal

UC-CPG-004 — Product Recall — Emergency Batch Identification and Withdrawal

Domain: Physical Goods & CPG **Actors:** QA Manager, Quality Director, Regulatory Affairs, Distribution Manager, Customer Service

Trigger: A quality defect is identified in a distributed product that requires a consumer-level recall.

Outcome: All affected batches identified in minutes. Recall initiated. Regulatory notified within required window. Distribution suspended. Consumer communication dispatched.

Step	Actor	Action	System Response	Event Published
1	QA Manager	Reports recall trigger in portal: defect type, severity, implicated batch range or date range.	Emergency CCR auto-created (type = Recall). Platform immediately queries batch provenance records.	CCR_SUBMITTED (emergency, Recall type)
2	Platform	—	Affected batch query executed: identifies all batches manufactured within the implicated range linked to the affected formulation version. Returns affected	RECALL_SCOPE_IDENTIFIED

			batch IDs, lot numbers, and distribution destinations in < 5 minutes.	
3	Quality Director	Reviews affected batch list. Confirms recall scope. Approves recall initiation.	Recall state → Active. All distribution records for affected batches flagged.	CCR_APPROVED (Recall)
4	Platform	—	Distribution suspension: all active shipments of affected batches flagged in distribution system. Carrier partners notified via logistics API.	DISTRIBUTION_SUSPENDED
5	Regulatory Affairs Manager	Submits recall notification to regulatory authority (FDA, EFSA, country-specific). Records submission reference and timing.	Regulatory notification artefact stored. Audit evidence of regulatory contact timing preserved.	COMPLIANCE_EVENTS_PUBLISHED
6	Distribution Manager	Coordinates physical product retrieval from retail locations. Updates retrieval progress in portal: batches recovered, quantities, destruction certificates.	Recall progress tracked in real time. Recall completion percentage calculated.	DISTRIBUTION_EVENTS_PUBLISHED (recall)
7	QA Manager	Records root cause analysis findings. Initiates CAPA (Corrective and Preventive Action) in platform.	CAPA linked to the Recall CCR. Formulation or process change CCR raised to address root cause.	INCIDENT_CLOSED

3.5 UC-CPG-005: Product Line Governance — Component Change Propagation

UC-CPG-005 — Product Line Change Governance — Component Update Propagation

Domain: Physical Goods & CPG **Actors:** Product Line Manager, Formulation Chemist, QA Manager, Regulatory Affairs

Trigger: A packaging material specification is updated, affecting 23 SKUs in a product line.

Outcome: All affected SKUs updated with new packaging specification. Each updated with provenance trace to the originating component change CCR.

Step	Actor	Action	System Response	Event Published
1	Product Line Manager	Raises CCR against the product line packaging component:	Platform identifies all 23 SKUs with a dependency on this packaging component.	CCR_SUBMITTED

		new sustainable packaging material, updated dimensions, revised printing specification.		
2	Platform	—	Cascade impact report generated: 23 affected SKUs across 4 product sub-lines. 2 SKUs require regulatory notification (label change). 21 SKUs require internal update only.	IMPACT_ANALYSIS_COMPLETE
3	Product Line Manager	Reviews impact report. Routes the 2 regulatory-notification SKUs to Regulatory Affairs. Approves the 21 internal-only updates.	CCR split into two tracks. 21 SKUs approved for immediate update. 2 SKUs pending regulatory assessment.	CCR_APPROVED (21 SKUs)
4	Formulation Chemist	Updates packaging specification for 21 SKUs. Each update versioned individually.	21 new specification versions created. Each linked to this CCR.	ARTEFACT_PUBLISHED (×21)
5	Regulatory Affairs Manager	Completes regulatory assessment for 2 SKUs. Submits label change notifications.	Notifications recorded. Both SKUs approved after regulatory acknowledgement (30-day window).	REGULATORY_GATE_PASSED (2 SKUs)
6	Formulation Chemist	Updates remaining 2 SKUs after regulatory approval.	All 23 SKUs updated. Product Line specification version bumped.	SOLUTION_VERSION_RELEASED

3.6 UC-CPG-006: Supply Chain Sustainability Provenance Audit

UC-CPG-006 — Supply Chain Sustainability Provenance Audit

Domain: Physical Goods & CPG **Actors:** Sustainability Manager, Supply Chain Director, Regulatory Affairs

Trigger: A major retail customer requests verified sustainability provenance for a product range for EU Deforestation Regulation compliance.

Outcome: Verified sustainability provenance report generated from platform records. All claims traceable to supply chain event records stored in platform.

Step	Actor	Action	System Response	Event Published
1	Sustainability Manager	Initiates provenance audit request in portal: product range, sustainability claim type (deforestation-free, palm oil traceability), target report date.	Platform queries artefact and distribution records for all batches in the product range.	AUDIT_INITIATED

2	Platform	—	Collates: raw material origin records, supplier certifications, processing event records, and distribution chain events from all ingested supply chain events.	COMPLIANCE_EVIDENCE_GATHERED
3	Supply Chain Director	Reviews provenance data. Identifies 3 supplier records with incomplete certification documentation.	Gaps flagged. Supplier data collection workflow initiated. Suppliers notified to upload updated certifications.	COMPLIANCE_GAPS_IDENTIFIED
4	Supply Chain Director	Suppliers upload updated certifications. Records added to platform artefact registry.	Certification artefacts stored. Provenance chain gaps resolved.	ARTEFACT_PUBLISHED (certifications)
5	Platform	—	Provenance report generated: complete supply chain trace for all products in scope. Report exported in EU Deforestation Regulation compliant format.	COMPLIANCE_REPORT_GENERATED
6	Sustainability Manager	Reviews and approves report. Sends to retail customer.	Report delivery recorded in audit trail. Regulatory evidence of compliance submission preserved.	AUDIT_COMPLETED

3.7 UC-CPG-007: New Capital Equipment — Commissioning and Lifecycle Registration

UC-CPG-007 — Capital Equipment Commissioning and Lifecycle Registration

Domain: Physical Goods & Manufacturing **Actors:** Manufacturing Engineer, QA Manager, Maintenance Manager

Trigger: A new CNC machine arrives on the factory floor and must be registered, commissioned, and brought into governed lifecycle management.

Outcome: Equipment registered as a Capital Good solution record. Commissioning qualification documented. Maintenance schedule active. Calibration records versioned.

Step	Actor	Action	System Response	Event Published
1	Manufacturing Engineer	Creates solution record: type = Capital Good. Enters equipment details: manufacturer, model, serial number, installation location, intended use.	Platform creates solution record. Lifecycle state → Commissioning. Equipment ID generated and linked to asset tag.	SOLUTION_CREATED

2	Manufacturing Engineer	Records Installation Qualification (IQ): confirms equipment installed per manufacturer specification. Attaches IQ documentation.	IQ artefact stored and linked to solution record.	ARTEFACT_PUBLISHED (IQ)
3	QA Manager	Conducts Operational Qualification (OQ): verifies equipment operates within defined parameters. Records test results.	OQ artefact stored. Quality gate evaluated.	QUALITY_GATE_PASSED
4	QA Manager	Conducts Performance Qualification (PQ): verifies equipment produces output meeting specification under production conditions.	PQ artefact stored. All qualification stages complete. Equipment lifecycle state → Active.	SOLUTION_STATE_CHANGED → Active
5	Maintenance Manager	Sets up preventive maintenance schedule in platform: calibration intervals, inspection dates, mandatory service events.	Maintenance schedule version-controlled. First scheduled calibration event created.	CONFIGURATION_VERSIONED
6	Maintenance Manager	At each calibration event: records calibration results. Calibration certificate stored as versioned artefact.	Equipment calibration record updated. Any out-of-specification finding triggers a defect and equipment lifecycle state → Under Review.	QUALITY_EVENTS_PUBLISHED

4 Financial Solutions Workflows

Use cases covering the lifecycle of financial solutions: investment funds, financial instruments, trading algorithms, insurance products, and capital solutions. Every product change is governed; every parameter update is tracked; every regulatory obligation is embedded in the workflow.

4.1 UC-FIN-001: Investment Fund Inception and Mandate Governance

UC-FIN-001 — Investment Fund Inception and Mandate Governance

Domain: Financial Solutions **Actors:** Portfolio Manager, Risk Manager, Compliance Officer, Legal, Investment Committee

Trigger: Investment team proposes to launch a new equity fund with a defined investment mandate.

Outcome: Fund solution record created. Mandate specification v1.0 approved by investment committee. Fund incepted with full governance audit trail.

Step	Actor	Action	System Response	Event Published
1	Portfolio Manager	Creates new solution record: type = Investment Solution. Enters mandate: asset class, benchmark, geographic scope, ESG constraints, concentration limits.	Platform creates solution record. Mandate document created as versioned artefact. Lifecycle state → In Development.	SOLUTION_CREATED
2	Risk Manager	Reviews mandate against risk framework. Runs risk model validation: factor exposure analysis, liquidity stress test, drawdown simulation.	Risk model validation results stored as artefact. Risk gate: all limits within approved bounds.	QUALITY_GATE_PASSED (Risk)
3	Compliance Officer	Reviews mandate for regulatory compliance: MiFID II product governance, UCITS eligibility, KID requirements (PRIIPs), AML screening.	Compliance gate evaluated. MiFID II product governance documentation generated.	COMPLIANCE_GATE_PASSED
4	Legal	Reviews fund documentation: prospectus, trust deed, subscription agreement. Records legal sign-off.	Legal approval recorded as artefact. Approval identity and timestamp preserved.	APPROVAL_RECORDED (Legal)
5	Investment Committee	Convenes to review full fund package. Votes on inception approval.	Approval recorded with all committee member identities and vote. Quorum confirmed.	APPROVAL_RECORDED (IC)
6	Portfolio Manager	Initiates fund inception in portal: sets fund	Fund lifecycle state → Active. Inception artefact created:	SOLUTION_STATE_CHANGED → Active

		inception date, registers with administrator, links to custody arrangements.	inception date, initial NAV, regulatory registration references.	
7	Platform	—	Inception event published. Fund available for investment. Mandate v1.0 locked. Any future mandate change requires a new CCR with IC approval.	SOLUTION_RELEASED → DISTRIBUTION_EVENTS_PUBLISHED

4.2 UC-FIN-002: Investment Mandate Change Governance

UC-FIN-002 — Investment Mandate Change — Governed Amendment

Domain: Financial Solutions **Actors:** Portfolio Manager, Risk Manager, Compliance Officer, Investment Committee

Trigger: Portfolio Manager proposes to expand the fund's geographic scope to include Emerging Markets.

Outcome: Mandate amended with full IC approval. All regulatory notifications dispatched. Investor disclosure updated. Mandate v2.0 locked.

Step	Actor	Action	System Response	Event Published
1	Portfolio Manager	Raises CCR: type = Mandate Amendment. Proposes: expand geographic scope to include Emerging Markets. Attaches investment rationale and risk analysis.	Platform computes risk score: Mandate Amendment = High Risk. Routes to Risk, Compliance, and IC.	CCR_SUBMITTED
2	Risk Manager	Runs updated risk model with EM expansion. Validates concentration limits, currency risk, liquidity profile still within bounds.	Risk validation artefact updated. Risk gate re-evaluated.	QUALITY_GATE_REEVALUATED
3	Compliance Officer	Reviews regulatory implications: updated UCITS stress test scenarios, revised KID risk indicator, investor notification requirement (31-day notice period).	Compliance obligations recorded. Investor notification window set.	COMPLIANCE_CHECK_PASSED
4	Investment Committee	Reviews amendment at scheduled IC meeting. Approves with IC quorum.	CCR state → Approved. Mandate v2.0 prepared.	CCR_APPROVED

5	Platform	—	Investor notification dispatched via distribution channel with 31-day notice of material mandate change.	CONSUMER_NOTIFICATIONS_SENT
6	Portfolio Manager	At end of notice period: confirms no material redemptions. Activates new mandate.	Mandate v2.0 locked in Artefact Registry. Fund configuration updated. Mandate v1.0 archived (still queryable for audit).	SOLUTION_VERSION_RELEASED

4.3 UC-FIN-003: Trading Algorithm — Research to Production

UC-FIN-003 — Trading Algorithm — Research SDE to Production Deployment

Domain: Financial Solutions **Actors:** Quant Researcher, Risk Manager, Technology Lead, Compliance Officer

Trigger: A quantitative researcher has developed a new equity momentum strategy and wants to deploy it to live trading.

Outcome: Algorithm approved through rigorous backtesting, risk validation, compliance review, and paper trading before live deployment. Full provenance from research to production.

Step	Actor	Action	System Response	Event Published
1	Quant Researcher	Creates solution record: type = Financial Instrument (Trading Algorithm). Provisions Financial Research SDE with quantitative libraries, backtesting framework, and historical data access.	Research SDE provisioned. Historical dataset pointers registered in Artefact Registry with checksums.	SOLUTION_CREATED → SDE_CREATED
2	Quant Researcher	Develops strategy in Research SDE. Each significant model version committed: hyperparameters, feature set, training data version recorded.	Each model version stored as artefact with: algorithm code hash, dataset version, hyperparameters, training metadata.	ARTEFACT_PUBLISHED (per model version)
3	Quant Researcher	Runs backtesting pipeline: executes strategy against historical market data in hermetic backtesting environment.	Backtest results stored as artefact: Sharpe ratio, max drawdown, VaR, turnover, transaction cost model. Build attestation generated.	ATTESTATION_CREATED (backtest)
4	Risk Manager	Reviews backtesting results. Validates against risk limits: Sharpe > 1.0, max drawdown < 20%, VaR	Risk validation gate evaluated. Risk artefact stored.	QUALITY_GATE_PASSED (Risk Validation)

		within limits. Runs independent stress test scenarios.		
5	Compliance Officer	Reviews for regulatory compliance: MAR market abuse risk assessment, MiFID II algorithmic trading requirements, pre-trade risk controls.	Compliance gate evaluated. MAR assessment stored as artefact.	COMPLIANCE_GATE_PASSED
6	Technology Lead	Deploys algorithm to paper trading environment (simulation stage): live market data feeds, production-equivalent infrastructure, but no real capital at risk.	Paper trading stage active. Real-time performance vs. backtest compared over 30-day period.	DEPLOYMENT_STARTED (paper_trading)
7	Risk Manager	Reviews paper trading results after 30 days. Performance within acceptable variance of backtest.	Paper trading gate passed. Algorithm approved for live deployment.	QUALITY_GATE_PASSED (Paper Trading)
8	Technology Lead	Deploys algorithm to live trading with initial capital allocation defined in mandate.	Algorithm live. Position limits enforced by pre-trade risk controls registered in platform.	DEPLOYED (live)

4.4 UC-FIN-004: Financial Model Drift Detection and Recalibration

UC-FIN-004 — Financial Model Drift Detection and Recalibration

Domain: Financial Solutions **Actors:** AI Agent, Risk Manager, Quant Researcher, Model Validation Committee

Trigger: AI Agent detects that a live pricing model's outputs are drifting from historical performance benchmarks.

Outcome: Model drift confirmed. Recalibration completed through governed CCR workflow. New model version deployed. Audit trail complete.

Step	Actor	Action	System Response	Event Published
1	AI Agent	—	Continuous monitoring of model output metrics. Detects: pricing model predictions deviating from realised prices by more than 2 standard deviations over 5-day window.	ANOMALY_DETECTED (model_drift)
2	AI Agent	—	Drift alert published: model ID, drift metric, comparison vs. historical baseline, confidence interval, recommended action.	AI_RECOMMENDATION_PUBLISHED

3	Risk Manager	Reviews drift alert. Confirms drift is material. Initiates model recalibration review.	CCR raised: type = Model Recalibration. Risk = High.	CCR_SUBMITTED
4	Quant Researcher	Investigates drift cause in Research SDE. Identifies regime change in underlying factor. Develops recalibrated model version.	New model version created in Research SDE. Recalibration artefact with updated parameters stored.	ARTEFACT_PUBLISHED (recalibrated_model)
5	Quant Researcher	Runs backtesting on recalibrated model. Validates improved fit to recent market regime.	Backtest results stored. Build attestation for recalibrated model generated.	ATTESTATION_CREATED
6	Model Validation Committee	Convenes. Reviews original model, drift evidence, recalibrated model backtesting. Approves deployment.	CCR approved. Model Validation artefact stored with committee sign-off.	CCR_APPROVED
7	Technology Lead	Deploys recalibrated model version through paper trading stage before live deployment.	Paper trading gate passed. Recalibrated model deployed to live.	DEPLOYED

4.5 UC-FIN-005: Tax Solution — Legislative Update to Effective-Date Release

UC-FIN-005 — Tax Solution — Legislative Update to Effective-Date Release

Domain: Financial Solutions / Legal & Tax **Actors:** Tax Product Manager, Tax Technologist, Legal, Compliance Officer

Trigger: New tax legislation is enacted with an effective date 60 days in the future. Tax product must be updated to comply.

Outcome: Tax rule set updated, validated across all affected jurisdictions, and released to activate automatically on the legislative effective date.

Step	Actor	Action	System Response	Event Published
1	Tax Product Manager	Creates CCR: type = Legislative Update. Attaches legislation reference, effective date, affected jurisdictions, change summary.	CCR automatically flagged as Regulatory-Triggered. Risk = High (legislative compliance mandatory). Routed to Legal and Compliance.	CCR_SUBMITTED

2	Tax Technologist	Implements updated tax rule set in Tax Solution SDE: updates calculation logic, rate tables, exemption rules for all affected jurisdictions.	New rule set version created. Each jurisdiction variant tracked separately.	CONFIGURATION_VERSIONED
3	Tax Technologist	Runs comprehensive validation suite: unit tests for each rule change, regression tests against previous results, cross-jurisdiction consistency tests.	Test suite passes. Coverage at 98% for modified rules.	TEST_SUITE_COMPLETED → QUALITY_GATE_PASSED
4	Legal	Reviews updated rule set for legal accuracy. Confirms correct interpretation of legislation.	Legal approval recorded with attorney identity.	APPROVAL_RECORDED (Legal)
5	Compliance Officer	Reviews compliance gate: confirms updated rule set will meet regulatory obligation on effective date.	Compliance gate passed. Effective date recorded in release record.	COMPLIANCE_GATE_PASSED
6	Tax Product Manager	Creates release: type = Effective-Date Release. Sets activation timestamp = legislative effective date at 00:01 local time per jurisdiction.	Release record created. All gates passed. Release state → Approved, Pending Effective Date.	RELEASE_APPROVED (pending)
7	Platform	—	At effective date timestamp: release automatically activated. New rule set deployed to all subscribed consumers simultaneously.	DEPLOYED (effective_date_activated)
8	Platform	—	Consumer notifications dispatched confirming rule set activation. Activation event published to DISTRIBUTION_EVENTS.	DISTRIBUTION_EVENTS_PUBLISHED

4.6 UC-FIN-006: Regulatory Examination Evidence Package

UC-FIN-006 — Regulatory Examination — Automated Evidence Package Generation

Domain: Financial Solutions **Actors:** Compliance Officer, Platform (automated), Regulator (read-only access)

Trigger: A regulatory authority (e.g. FCA) notifies the firm of an upcoming examination covering product governance and model risk management.

Outcome: Complete evidence package generated from platform records in hours, not weeks. Examination conducted with full documentation available.

Step	Actor	Action	System Response	Event Published
1	Compliance Officer	Receives examination notice. Defines scope in platform: regulatory framework = MiFID II Product Governance, time period = 24 months, products in scope = all live investment solutions.	Platform queues evidence generation for defined scope.	AUDIT_INITIATED
2	Platform	—	Evidence collection: queries all CCRs, mandate amendments, model validation records, IC approvals, risk gate results, compliance check events for all in-scope solutions over the defined period.	COMPLIANCE_EVIDENCE_GATHERED
3	Platform	—	Compliance report generated: structured by MiFID II control requirements. Each control mapped to specific platform evidence records with timestamps and actor identities.	COMPLIANCE_REPORT_GENERATED
4	Compliance Officer	Reviews generated report. Identifies 3 controls with partial manual evidence. Uploads supplementary documents.	Manual evidence stored as artefacts linked to relevant controls.	ARTEFACT_PUBLISHED (manual_evidence)
5	Compliance Officer	Optionally activates Regulator Access portal: time-limited read-only access for examining regulator to query platform audit trail directly.	Regulator Access session created. All regulator access actions logged in audit trail.	REGULATOR_ACCESS_GRANTED
6	Compliance Officer	Examination completed. Regulator Access session revoked.	Session revocation recorded. Complete examination interaction preserved in audit trail.	AUDIT_COMPLETED

4.7 UC-FIN-007: Insurance Product Launch and Term Governance

UC-FIN-007 — Insurance Product Launch and Policy Term Governance

Domain: Financial Solutions **Actors:** Product Actuary, Compliance Officer, Legal, Distribution Manager

Trigger: Insurance firm designs and launches a new term life insurance product.

Outcome: Product terms v1.0 approved. Actuarial model validated. Product launched across all distribution channels. All term changes governed by CCR.

Step	Actor	Action	System Response	Event Published
1	Product Actuary	Creates solution record: type = Insurance Product. Enters product parameters: coverage amounts, premium rates, exclusions, underwriting criteria.	Solution record created. Policy terms stored as versioned artefact.	SOLUTION_CREATED
2	Product Actuary	Runs actuarial model validation: mortality tables, pricing adequacy analysis, capital requirement calculation (Solvency II standard formula).	Actuarial validation artefact stored. Capital adequacy gate passed.	QUALITY_GATE_PASSED (Actuarial)
3	Compliance Officer	Reviews product for regulatory compliance: FCA Consumer Duty requirements, ICOBs rules, fair value assessment, claims handling procedures.	Compliance gate passed. Fair value assessment document stored.	COMPLIANCE_GATE_PASSED
4	Legal	Drafts and approves policy wording. Records sign-off.	Policy wording artefact stored with legal approval.	APPROVAL_RECORDED
5	Distribution Manager	Configures distribution: broker network, direct channel, bancassurance partner.	Distribution configuration version-controlled.	CONFIGURATION_VERSIONED
6	Product Actuary	Initiates product launch.	Release gate evaluated. All gates passed. Product state → Active. Policy terms v1.0 locked.	SOLUTION_RELEASED
7	Platform	—	Any future change to premium rates, exclusions, or underwriting criteria requires a CCR with actuarial and compliance review, ensuring the product remains fairly valued and compliant throughout its lifetime.	CONFIGURATION_DRIFT_PROTECTED

5 Agricultural & Agribusiness Workflows

Use cases for agricultural solutions: precision farming platforms, crop management systems, supply chain traceability, IoT device lifecycle, livestock management, and agribusiness marketplace platforms.

5.1 UC-AGRI-001: Agricultural IoT Platform — Onboarding and Governed Deployment

UC-AGRI-001 — Agricultural IoT Platform — Onboarding and Governed Deployment

Domain: Agricultural & Agribusiness **Actors:** AgTech Engineer, Farm Operations Manager, Platform Admin

Trigger: A new precision irrigation IoT platform is registered and deployed to a farm network.

Outcome: IoT platform solution record created. Edge SDE provisioned. Firmware pipeline operational. All field devices registered and enrolled.

Step	Actor	Action	System Response	Event Published
1	AgTech Engineer	Creates solution record: type = Agricultural Solution. Registers IoT platform with device types, firmware architecture, target deployment (farm network, 150 devices).	Solution record created. Edge SDE provisioned with cross-compilation toolchain and pre-populated Nix dependency cache for offline operation.	SOLUTION_CREATED → SDE_CREATED
2	AgTech Engineer	Develops firmware in Edge SDE. Each firmware version committed and built hermetically.	Hermetic cross-compile build produces firmware image. SHA-256 digest generated. SLSA-equivalent attestation created for firmware artefact.	BUILD_PASSED → ATTESTATION_CREATED
3	AgTech Engineer	Runs firmware in IoT device simulation environment: validates command/control, sensor data telemetry, offline operation queue behaviour.	Simulation test suite passes. Firmware quality gate evaluated.	QUALITY_GATE_PASSED
4	Farm Operations Manager	Reviews firmware release notes. Approves deployment to farm devices.	Release approved. OTA update pipeline activated.	RELEASE_APPROVED
5	Platform	—	OTA firmware update deployed to all 150 field devices via secure update channel. Each device reports update confirmation with device ID and installed firmware digest.	DISTRIBUTION_EVENTS_PUBLISHED (per device)

6	Platform	—	All device update confirmations received. Distribution record complete: 150 devices enrolled, firmware version attested, deployment timestamp recorded.	DEPLOYMENT_COMPLETE
7	Platform	—	IoT telemetry stream active: sensor data flows from field devices to platform event bus. AI Agent begins baseline learning for anomaly detection and yield modelling.	AI_MONITORING_ACTIVE

5.2 UC-AGRI-002: Crop Management — Seasonal Release Cycle

UC-AGRI-002 — Crop Management System — Seasonal Release Cycle

Domain: Agricultural & Agribusiness **Actors:** AgTech Engineer, Agronomist, Farm Operations Manager, QA Engineer

Trigger: Spring planting season approaches. Crop management system must be updated with new season's agronomic rule set and AI model.

Outcome: Seasonal release deployed and activated before planting season start date. New agronomic rules and AI yield model active across all enrolled farms.

Step	Actor	Action	System Response	Event Published
1	Agronomist	Raises CCR: type = Agronomic Rule Update. Attaches updated crop calendar, variety-specific parameters, pest management thresholds, irrigation scheduling rules for the upcoming season.	CCR submitted. Risk score = Medium (seasonal update, well-defined scope). Routes to Farm Operations Manager for approval.	CCR_SUBMITTED
2	AgTech Engineer	Updates agronomic rule set in SDE. Updates AI yield model with new season's training data (prior season outcomes, weather patterns, soil data).	New rule set and model version created. Training dataset version linked to model artefact.	ARTEFACT_PUBLISHED
3	QA Engineer	Runs validation suite: unit tests for agronomic rule logic, backtesting of AI model against prior season data, simulation of irrigation	Test suite passes. AI model performance metrics within approved bounds.	QUALITY_GATE_PASSED

		scheduling against historical weather.		
4	Farm Operations Manager	Reviews seasonal update. Approves seasonal release.	CCR approved. Release created with seasonal effective date = planting season start minus 5 days.	RELEASE_APPROVED (seasonal effective date)
5	Platform	—	At seasonal effective date: release activates automatically. All enrolled farm devices receive updated rule set and AI model via OTA.	DEPLOYED (seasonal_activation)
6	AI Agent	—	Planting season monitoring begins: real-time yield predictions issued per field, irrigation recommendations generated, pest risk alerts published.	AI_RECOMMENDATION_PUBLISHED (per field, per event)

5.3 UC-AGRI-003: Farm-to-Consumer Traceability Query

UC-AGRI-003 — Farm-to-Consumer Traceability — End-to-End Provenance Query

Domain: Agricultural & Agribusiness **Actors:** Retailer Compliance Team, Qala Platform, Food Safety Regulator

Trigger: A retailer receives a food safety alert from a regulator about a produce lot and needs to trace origin within 2 hours.

Outcome: Complete provenance chain from consumer product → distribution → processing → farm → field identified in under 10 minutes.

Step	Actor	Action	System Response	Event Published
1	Retailer Compliance	Enters product lot code into Qala traceability query: lot number, product type, approximate date.	Platform queries distribution records, batch records, and supply chain event chain using lot number as primary key.	TRACEABILITY_QUERY_INITIATED
2	Platform	—	Queries distribution record: identifies processing facility, processing date, and input batch IDs.	DISTRIBUTION_RECORD_FOUND
3	Platform	—	Queries batch records at processing facility (ingested from MES via integration): identifies raw produce input lots, arrival dates, cold chain event history.	BATCH_RECORDS_FOUND
4	Platform	—	Queries farm supply chain events: identifies originating farms, field IDs, harvest dates,	PROVENANCE_CHAIN_COMPLETE

			harvest method. GS1 EPCIS events traced to farm.	
5	Platform	—	Complete provenance chain assembled: consumer lot → 3 processing input lots → 2 originating farms → 4 specific fields → harvest dates → agronomic treatment records.	TRACEABILITY_REPORT_GENERATED
6	Retailer Compliance	Reviews provenance report in < 10 minutes. Shares with food safety regulator.	Report exported. Regulatory response initiated based on identified farm scope.	AUDIT_COMPLETED
7	Platform	—	All queries logged in immutable audit trail. Retailer access to provenance data recorded.	AUDIT_EVENTS_PUBLISHED

This workflow demonstrates that Qala's batch and distribution event architecture enables recall scope identification and full provenance tracing in under 10 minutes — compared to the 2–7 day industry average for manual traceability investigations.

5.4 UC-AGRI-004: AI-Driven Yield Optimisation Recommendation

UC-AGRI-004 — AI-Driven Yield Optimisation — Field-Level Recommendations

Domain: Agricultural & Agribusiness **Actors:** AI Agent, Agronomist, Farm Operations Manager

Trigger: AI Agent analyses mid-season sensor data and identifies yield optimisation opportunity.

Outcome: Agronomist reviews AI recommendation. Targeted intervention applied. Yield improvement measured against baseline.

Step	Actor	Action	System Response	Event Published
1	AI Agent	—	Analyses 30-day sensor data stream: soil moisture variance across field sections, canopy temperature anomalies, growth stage progression vs. expected curve.	AI_ANALYSIS_RUNNING
2	AI Agent	—	Detects: Field Sections 3 and 7 showing moisture deficit pattern correlated with 12% yield reduction in historical similar conditions. Generates irrigation intervention recommendation with 3 scenario options.	AI_RECOMMENDATION_PUBLISHED
3	Agronomist	Reviews recommendation in portal. Reviews	—	

		supporting evidence: sensor data charts, historical comparison, 3 intervention scenarios with predicted yield impact.		
4	Agronomist	Accepts Scenario B: targeted drip irrigation increase in Sections 3 and 7 over 14 days. Confirms acceptance in portal.	Recommendation accepted. Feedback recorded for model learning. Irrigation system CCR auto-generated for the configuration change.	AI_FEEDBACK_RECORDED
5	Platform	—	Irrigation controller configuration updated via OTA with new schedule for Sections 3 and 7. Change logged against the CCR.	DEPLOYED (irrigation_config_update)
6	AI Agent	—	Post-intervention monitoring: tracks moisture levels, growth response, compares to predicted trajectory. Issues updated yield forecast after 7 days.	AI_RECOMMENDATION_PUBLISHED (progress_update)
7	Farm Operations Manager	Reviews end-of-season yield actuals. Sections 3 and 7 yield 11% above baseline — within 1% of AI prediction.	Outcome recorded in platform. AI model performance metric updated. Agronomist outcome data feeds next season model training.	QUALITY_EVENTS_PUBLISHED (outcome)

5.5 UC-AGRI-005: Agricultural Input Product — From Regulatory Approval to Farm Distribution

UC-AGRI-005 — Agricultural Input Product — Regulatory Approval to Farm Distribution

Domain: Agricultural & Agribusiness **Actors:** Product Developer, Regulatory Affairs, Distribution Manager

Trigger: AgInput company develops a new bio-stimulant product and needs to bring it from R&D through regulatory approval to market distribution.

Outcome: Product approved by regulatory authorities (EPA/EFSA). Distributed to farm retailers and direct-to-farm channels with full traceability.

Step	Actor	Action	System Response	Event Published
1	Product Developer	Creates solution record: type = Agricultural Solution (Input Product). Registers product: bio-stimulant, active ingredient, target	Solution record created. Lab SDE provisioned for efficacy and safety study management.	SOLUTION_CREATED

		crops, application method.		
2	Product Developer	Records trial results from field and lab studies: efficacy data, environmental safety data, mammalian toxicology data. Imports from LIMS.	Study artefacts stored. Linked to product solution record.	ARTEFACT_PUBLISHED (study_data)
3	Regulatory Affairs Manager	Prepares regulatory dossier: compiles all study data, formulation specification, label draft, environmental risk assessment.	Platform generates regulatory dossier compilation artefact from stored records.	ARTEFACT_PUBLISHED (dossier)
4	Regulatory Affairs Manager	Submits to EPA (US) and EFSA (EU). Records submission references and expected approval timeline.	Regulatory gate status = Submitted (US and EU separately tracked).	REGULATORY_GATE_UPDATED
5	Regulatory Affairs Manager	Records EPA approval. Records EFSA approval (6 weeks later). Attaches approval certificates.	Both regulatory gates → Approved. Lifecycle state → Approved for Distribution.	SOLUTION_STATE_CHANGED
6	Distribution Manager	Configures distribution: farm retail channel, direct-to-farm subscription, distributor network per region.	Distribution configuration version-controlled. Regional compliance status per country recorded.	CONFIGURATION_VERSIONED
7	Distribution Manager	Releases product for distribution. Batch records imported from manufacturing partner.	Distribution events published. Product launched. Full traceability from formulation through distribution active.	DISTRIBUTION_EVENTS_PUBLISHED

5.6 UC-AGRI-006: Livestock Management System — Health Protocol Governance

UC-AGRI-006 — Livestock Management — Health Protocol Governance and Change Control

Domain: Agricultural & Agribusiness **Actors:** Veterinary Officer, Farm Operations Manager, Regulatory Affairs

Trigger: A new veterinary health protocol is required following updated regulatory guidance on antibiotic use.

Outcome: Updated health protocol version approved. All enrolled farms notified. Protocol change recorded with full audit trail. Compliance with new guidance confirmed.

Step	Actor	Action	System Response	Event Published

1	Veterinary Officer	Raises CCR: type = Health Protocol Update. Attaches regulatory guidance reference, proposed protocol changes: antibiotic selection criteria, withdrawal period updates, record-keeping requirements.	CCR submitted. Risk = High (regulatory compliance). Routes to Farm Operations Manager and Regulatory Affairs.	CCR_SUBMITTED
2	Regulatory Affairs Manager	Verifies protocol update correctly interprets new regulatory guidance. Provides compliance sign-off.	Compliance approval recorded.	APPROVAL_RECORDED
3	Farm Operations Manager	Reviews operational impact. Approves protocol update.	CCR approved.	CCR_APPROVED
4	Platform	—	New protocol version created. All enrolled farms in the livestock management system notified of protocol change with effective date and training materials.	CONSUMER_NOTIFICATIONS_SENT
5	Farm Operations Manager	Distributes updated protocol to farm managers. Training completion recorded in platform.	Training completion events recorded per farm per manager.	QUALITY_EVENTS_PUBLISHED (training)
6	Platform	—	Health protocol version update recorded in compliance audit trail. All subsequent health events logged against the new protocol version — traceability from animal health event to protocol version to regulatory guidance.	COMPLIANCE_EVENTS_PUBLISHED

6 Professional Services & Consulting Workflows

Use cases for service solutions: consulting methodology governance, managed service lifecycle, service catalogue management, engagement delivery, and professional service quality management.

6.1 UC-SVC-001: Consulting Methodology — Versioning and Knowledge Governance

UC-SVC-001 — Consulting Methodology — Versioning and Knowledge Governance

Domain: Professional Services & Consulting **Actors:** Partner, Principal Consultant, Knowledge Manager

Trigger: A Partner wants to formalise and version a proprietary methodology developed across 12 recent engagements.

Outcome: Methodology v1.0 published as a versioned, searchable solution record. All future updates governed by CCR. Intellectual capital preserved and accessible to all consultants.

Step	Actor	Action	System Response	Event Published
1	Principal Consultant	Creates solution record: type = Consulting Service (Methodology). Provisions Knowledge SDE with document management and collaborative editing tools.	Solution record created. Knowledge SDE activated. Initial methodology version 0.1 created.	SOLUTION_CREATED → SDE_CREATED
2	Principal Consultant	Structures methodology content: phases, activities, tools, templates, case studies, common pitfalls. Each component version-controlled as an artefact.	All methodology components stored as versioned artefacts: documents, templates, frameworks, playbooks.	ARTEFACT_PUBLISHED (per component)
3	Partner	Reviews methodology completeness and quality. Provides editorial sign-off.	Partner approval recorded.	APPROVAL_RECORDED
4	Knowledge Manager	Configures AI indexing: embeds all methodology components into the knowledge graph for semantic search and retrieval.	AI Agent knowledge graph updated. Methodology searchable by concept, industry, engagement phase.	AI_INDEXING_COMPLETE
5	Platform	—	Methodology v1.0 published to internal service catalogue.	SOLUTION_RELEASED →

			All consultants notified of new approved methodology.	DISTRIBUTION_EVENTS_PUBLISHED
6	Consultant	Queries knowledge base: "digital transformation operating model, retail sector, phase 2 stakeholder engagement."	AI Agent returns: exact methodology section with relevance score, 3 similar past engagement references, recommended templates.	AI_RECOMMENDATION_PUBLISHED
7	Principal Consultant	Following new engagement learnings: raises CCR to update methodology. New version reviewed, approved, published.	Methodology evolves under governance. Every version traceable. Knowledge compounds rather than evaporates.	CCR_APPROVED → SOLUTION_VERSION_RELEASED

6.2 UC-SVC-002: Managed Service — SLA Definition and Governed Change

UC-SVC-002 — Managed Service — SLA Definition and Governed SLA Change

Domain: Professional Services & Consulting **Actors:** Service Architect, Account Manager, Legal, Compliance Officer, Customer

Trigger: A managed service provider designs a new managed cloud operations service and governs a subsequent SLA upgrade request.

Outcome: Service specification v1.0 published. SLA upgrade governed through CCR. Customer notified. New SLA v2.0 activated.

Step	Actor	Action	System Response	Event Published
1	Service Architect	Creates solution record: type = Managed Service. Defines service specification: scope, SLA tiers (response time, uptime target, escalation path), runbooks, support hours.	Solution record created. Service specification v1.0 stored as versioned artefact.	SOLUTION_CREATED
2	Legal	Reviews service specification. Confirms contractual language aligns with service terms.	Legal approval recorded.	APPROVAL_RECORDED
3	Service Architect	Publishes service to service catalogue.	Service specification v1.0 locked. Service state → Active.	SOLUTION_RELEASED
4	Account Manager	Customer requests SLA upgrade: uptime from 99.5% to 99.9%.	CCR submitted: type = SLA Amendment. Risk = Medium. Routes to Service Architect and Legal.	CCR_SUBMITTED

		Raises CCR on behalf of customer.		
5	Service Architect	Reviews technical feasibility and cost model. Approves SLA amendment.	CCR approved. Service specification v2.0 drafted.	CCR_APPROVED
6	Legal	Reviews updated contractual SLA terms.	Legal approval recorded for v2.0.	APPROVAL_RECORDED
7	Platform	—	Service specification v2.0 activated. Customer notified. Old specification v1.0 archived but queryable.	SOLUTION_VERSION_RELEASED → CONSUMER_NOTIFICATIONS_SENT

6.3 UC-SVC-003: Engagement Delivery — Deliverable Governance and Outcome Tracking

UC-SVC-003 — Consulting Engagement Delivery — Deliverable Governance and Outcomes

Domain: Professional Services & Consulting **Actors:** Engagement Manager, Consultants, Client, Quality Manager

Trigger: A consulting engagement is underway. A major deliverable (operating model design) must be governed, client-approved, and outcome-tracked.

Outcome: Deliverable v1.0 approved by client. Outcome metrics defined. Post-engagement outcome results linked to engagement record for institutional learning.

Step	Actor	Action	System Response	Event Published
1	Engagement Manager	Creates engagement solution record. Links to methodology. Creates deliverable work item: type = Operating Model Design.	Engagement record created. Deliverable work item tracked in platform.	SOLUTION_CREATED (engagement)
2	Consultants	Develop deliverable in Knowledge SDE. Use approved methodology components as reference. Version each draft.	Deliverable versions stored as artefacts. Each draft traceable to specific methodology components used.	ARTEFACT_PUBLISHED (deliverable_drafts)
3	Quality Manager	Conducts internal quality review. Reviews against methodology quality checklist. Provides QA sign-off.	QA approval recorded. Quality gate passed.	QUALITY_GATE_PASSED
4	Engagement Manager	Submits deliverable to client for review.	Client review event recorded. Distribution record created.	DISTRIBUTION_EVENTS_PUBLISHED

5	Client	Reviews deliverable. Requests revisions on 2 sections. Records feedback in portal (or via engagement manager).	Revision requests recorded. Deliverable state → In Revision.	DEFECT_CREATED (revision_requests)
6	Consultants	Revise affected sections. Engagement Manager resubmits.	Deliverable v1.1 stored. Revision traceable to client feedback.	ARTEFACT_PUBLISHED (v1.1)
7	Client	Approves deliverable v1.1.	Client approval recorded with timestamp. Deliverable state → Approved. Engagement outcome metrics defined.	APPROVAL_RECORDED (client)
8	Platform	—	6-month post-engagement: outcome survey results ingested. Outcome metrics compared to engagement success criteria. Results stored and indexed by AI Agent for future methodology improvement.	QUALITY_EVENTS_PUBLISHED (outcomes)

6.4 UC-SVC-004: Service Catalogue Governance — New Service Approval and Publication

UC-SVC-004 — Service Catalogue Governance — New Service Approval and Publication

Domain: Professional Services & Consulting **Actors:** Service Design Lead, Legal, Sales Enablement, Marketing

Trigger: A new coaching programme is ready to be added to the organisation's external service catalogue.

Outcome: Coaching programme approved through service catalogue governance workflow. Published to external service catalogue. Sales team enabled.

Step	Actor	Action	System Response	Event Published
1	Service Design Lead	Creates solution record: type = Consulting Service (Coaching Programme). Defines programme: structure, session count, target persona, success criteria, pricing.	Solution record created. Programme artefacts (structure doc, session guides, assessment tools) stored.	SOLUTION_CREATED
2	Service Design Lead	Runs pilot programme with 3 beta clients. Records participant outcomes and feedback.	Pilot outcomes stored as quality events. NPS and goal completion rate calculated.	QUALITY_EVENTS_PUBLISHED (pilot)

3	Legal	Reviews service terms, IP ownership, confidentiality provisions for the coaching engagement format.	Legal approval recorded.	APPROVAL_RECORDED
4	Service Design Lead	Initiates service catalogue publication workflow.	Catalogue CCR generated: type = New Service Addition. Routes to Sales Enablement and Legal.	CCR_SUBMITTED (catalogue_add)
5	Sales Enablement	Reviews commercial viability, pricing positioning, competitive differentiation.	Sales approval recorded.	APPROVAL_RECORDED (Sales)
6	Platform	—	Coaching programme published to external service catalogue. Sales team notified. Programme materials accessible via AI-powered knowledge retrieval.	SOLUTION_RELEASED → DISTRIBUTION_EVENTS_PUBLISHED

6.5 UC-SVC-005: Subscription Service — Feature Release and Consumer Notification

UC-SVC-005 — Subscription Service — Feature Release and Subscriber Notification

Domain: Professional Services & Consulting **Actors:** Product Manager (Service), Engineering Lead, Customer Success

Trigger: A subscription-based advisory service is releasing a new AI-powered benchmarking feature to all subscribers.

Outcome: Feature released to all subscribers. Tiered rollout based on subscription tier. All subscribers notified. Feature adoption tracked.

Step	Actor	Action	System Response	Event Published
1	Product Manager	Creates release: new feature addition to Subscription Service solution. Defines rollout plan: Enterprise tier first (Day 1), Professional tier (Day 7), Standard tier (Day 14).	Release record created. Tiered distribution plan attached.	RELEASE_PLANNED
2	Engineering Lead	Verifies feature testing complete. All gate checks passed: functional test, performance test, compatibility with	Release gates passed.	QUALITY_GATE_PASSED

		existing subscriber configurations.		
3	Product Manager	Approves release.	Release state → Approved.	RELEASE_APPROVED
4	Platform	—	Day 1: Feature activated for all Enterprise subscribers. Activation event published.	DISTRIBUTION_EVENTS_PUBLISHED (enterprise_tier)
5	Customer Success	Monitors adoption metrics for Enterprise tier over 7 days. Collects feedback. No critical issues.	Adoption metrics published as quality events.	QUALITY_EVENTS_PUBLISHED
6	Platform	—	Day 7: Feature activated for Professional tier. Day 14: Standard tier. All subscribers eventually receive feature.	DISTRIBUTION_EVENTS_PUBLISHED (professional, standard)
7	Customer Success	Tracks feature adoption rate across all tiers. 78% adoption within 30 days.	Adoption metric recorded. Feature outcome linked to release record.	QUALITY_EVENTS_PUBLISHED (adoption_metric)

6.6 UC-SVC-006: Knowledge Retrieval — AI-Powered Methodology Search

UC-SVC-006 — AI Knowledge Retrieval — Cross-Engagement Institutional Learning

Domain: Professional Services & Consulting **Actors:** Consultant, AI Agent

Trigger: A junior consultant needs to rapidly prepare for a client meeting on supply chain resilience strategy for a manufacturing client.

Outcome: AI Agent retrieves and synthesises relevant methodologies, past engagement references, and recommended frameworks in under 2 minutes.

Step	Actor	Action	System Response	Event Published
1	Consultant	Queries knowledge base: "supply chain resilience framework, manufacturing sector, 6-month transformation programme."	AI Agent performs semantic search across: methodology library, approved frameworks, anonymised engagement references, approved case studies.	AI_ANALYSIS_STARTED
2	AI Agent	—	Returns top 5 results: (1) Supply Chain Resilience Methodology v3.2 — relevance 94%; (2) Transformation Roadmap Framework v2.0 — 87%; (3) 3 anonymised engagement summaries with outcome metrics; (4) Recommended assessment tool: Supply Chain Maturity Matrix.	AI_RECOMMENDATION_PUBLISHED

3	Consultant	Reviews recommendations. Opens Supply Chain Resilience Methodology. Navigates to Phase 2: Supplier Risk Assessment.	AI Agent tracks navigation. Logs content accessed for usage analytics.	AI_EVENTS_PUBLISHED
4	Consultant	Queries further: "supplier risk assessment, dual-sourcing decision framework, cost-risk trade-off."	AI Agent drills deeper into methodology. Returns specific section with relevant templates pre-loaded.	AI_RECOMMENDATION_PUBLISHED (deep_query)
5	Consultant	Downloads recommended templates. Rates AI results: 4 stars for relevance.	Feedback recorded. AI model receives positive reinforcement on this retrieval pattern.	AI_FEEDBACK_RECORDED
6	AI Agent	—	Post-engagement: flags that this engagement will generate new knowledge relevant to supply chain resilience. Suggests consultant submit key insights for methodology update review.	AI_RECOMMENDATION_PUBLISHED (contribution_suggestion)

7 Business, Process & Operational Solution Workflows

Use cases covering business solutions: process redesign governance, operational solution launch and lifecycle, workflow automation management, and post-launch drift detection.

7.1 UC-BIZ-001: Business Process Redesign — Governed Launch and Post-Launch Monitoring

UC-BIZ-001 — Business Process Redesign — Governed Launch and Drift Detection

Domain: Business & Operational **Actors:** Process Owner, Business Analyst, Operations Director, Change Champion, Impacted Teams

Trigger: An organisation designs a new order-to-cash process replacing an existing manual workflow.

Outcome: New process launched under governance. Post-launch drift from as-designed state detected and remediated within 30 days.

Step	Actor	Action	System Response	Event Published
1	Business Analyst	Creates solution record: type = Process Solution. Attaches BPMN process model v1.0, process metrics baseline, KPI targets.	Solution record created. Process model artefact version-controlled in artefact registry.	SOLUTION_CREATED
2	Process Owner	Raises CCR for process launch: attaches stakeholder impact assessment, training plan, rollout schedule.	CCR submitted. Risk = High (cross-functional impact). Routes to Operations Director and Impacted Team Leads.	CCR_SUBMITTED
3	Operations Director	Reviews impact assessment. Approves launch with phased rollout by region.	CCR approved. Launch schedule recorded.	CCR_APPROVED
4	Change Champion	Delivers training. Records completion per team.	Training completion events published. Readiness gate evaluated.	QUALITY_GATE_PASSED (readiness)
5	Process Owner	Launches process in Region 1. Monitors KPIs: cycle time, error rate, throughput.	Process state → Active (Region 1). Telemetry ingestion from operational system via Quality Event Ingestion API.	SOLUTION_RELEASED
6	Platform	—	30 days post-launch: AI Agent analyses operational telemetry. Detects cycle time 18% above designed target in one step due to manual	ANOMALY_DETECTED (process_drift)

			exception handling not captured in the original design.	
7	Business Analyst	Reviews AI analysis. Confirms drift. Raises CCR: type = Process Amendment. Updates BPMN model to capture the exception handling pathway explicitly.	CCR approved. Process model v1.1 published. Drift now the governed definition.	CCR_APPROVED → SOLUTION_VERSION_RELEASED

7.2 UC-BIZ-002: Digital Transformation Programme — Portfolio Governance

UC-BIZ-002 — Digital Transformation Programme — Multi-Solution Portfolio Governance

Domain: Business & Operational **Actors:** Programme Director, Initiative Owners, Executive Sponsor, PMO

Trigger: A digital transformation programme spanning 8 initiatives across 3 years needs a single governed view of all solution lifecycles.

Outcome: Unified portfolio view established. All 8 initiatives tracked as governed solutions. Cross-initiative dependencies managed. Programme outcomes measured against commitments.

Step	Actor	Action	System Response	Event Published
1	Programme Director	Creates Solution Factory for the transformation programme. Registers all 8 initiatives as solution records of appropriate types: Business Solution, Process Solution, Technology Platform.	Solution Factory created. 8 solution records created and enrolled. Programme-level governance policies applied.	SOLUTION_FACTORY_CREATED
2	Initiative Owners	Each owner provisions appropriate SDE and begins development work. Cross-initiative dependencies declared in solution relationship graph.	SDEs provisioned per initiative. Dependency graph populated.	SDE_CREATED (×8)
3	PMO	Configures programme-level quality dashboard: milestone completion, dependency health, cross-initiative risk indicators.	Dashboard active. Real-time status visible to all stakeholders.	CONFIGURATION_VERSIONED

4	Platform	—	AI Agent identifies: Initiative 5 dependency on Initiative 2 output is at risk (Initiative 2 is 3 weeks behind plan). Alert issued to Programme Director.	AI_RECOMMENDATION_PUBLISHED (dependency_risk)
5	Programme Director	Reviews alert. Convenes dependency resolution meeting. Approves accelerated Initiative 2 delivery plan.	Resolution plan recorded as CCR. Adjusted timeline documented.	CCR_APPROVED
6	Platform	—	Programme-level quality gate at quarterly milestone: all 8 initiatives assessed against committed outcomes. 6 on track, 2 flagged for Executive Sponsor review.	QUALITY_GATE_EVALUATED (programme_milestone)
7	Executive Sponsor	Reviews 2 flagged initiatives. Approves revised scope for one; provides additional resource for other.	Executive decisions recorded with rationale. Programme record updated.	APPROVAL_RECORDED (executive)

7.3 UC-BIZ-003: Workflow Automation — Build, Test, Deploy and Monitor

UC-BIZ-003 — Workflow Automation — Build, Govern, and Monitor an Operational Automation

Domain: Business & Operational **Actors:** Operations Engineer, Operations Director, QA Engineer

Trigger: Operations team builds a new automated invoice processing workflow to replace a manual process.

Outcome: Automation workflow built, tested, released, and monitored. Exception handling rules governed by versioned configuration.

Step	Actor	Action	System Response	Event Published
1	Operations Engineer	Creates solution record: type = Workflow Solution. Provisions SDE with workflow automation tooling (e.g. RPA, integration platform).	Solution record created. SDE provisioned.	SOLUTION_CREATED
2	Operations Engineer	Builds automation workflow. Commits each version. Each configuration change (exception rules, routing logic, approval thresholds) requires a CCR.	Workflow versions tracked. Configuration changes governed.	ARTEFACT_PUBLISHED (per version)

3	QA Engineer	Tests automation: unit tests for each business rule, end-to-end tests with representative data, exception scenario tests.	Test results published. Quality gate evaluated.	QUALITY_GATE_PASSED
4	Operations Director	Reviews automation against operational requirements. Approves production deployment.	Release approved.	RELEASE_APPROVED
5	Platform	—	Automation deployed. Operational telemetry streams into platform: processed invoice count, exception rate, cycle time.	DEPLOYED → MONITORING_ACTIVE
6	Platform	—	AI Agent detects: exception rate spiking on invoices with multi-currency amounts. Pattern identified and alert raised.	ANOMALY_DETECTED
7	Operations Engineer	Reviews alert. Identifies edge case in currency conversion logic. Raises CCR to update the exception handling rule.	CCR approved. New configuration version deployed.	CCR_APPROVED → DEPLOYED (config update)

7.4 UC-BIZ-004: Operational Policy — Version Control and Compliance Audit

UC-BIZ-004 — Operational Policy Governance — Version Control and Compliance Audit

Domain: Business & Operational **Actors:** Policy Owner, Compliance Manager, Impacted Teams

Trigger: An organisation's expense reimbursement policy must be updated to reflect new travel regulations.

Outcome: Policy updated under governance. All impacted employees notified. Compliance audit trail complete. Old policy version archived but auditable.

Step	Actor	Action	System Response	Event Published
1	Policy Owner	Creates CCR: type = Policy Amendment. Describes changes to expense limits, approval thresholds, and documentation requirements.	CCR submitted. Impacted policy document identified in artefact registry.	CCR_SUBMITTED
2	Compliance Manager	Reviews policy amendment for	Compliance review artefact stored.	APPROVAL_RECORDED (Compliance)

		regulatory compliance with updated travel and tax regulations.		
3	Policy Owner	Approves final policy wording. Approves CCR.	CCR approved. Policy document v2.0 created in artefact registry.	CCR_APPROVED
4	Platform	—	Policy v2.0 published. All impacted employees notified via notification service. Effective date confirmed.	SOLUTION_VERSION_RELEASED → CONSUMER_NOTIFICATIONS_SENT
5	Compliance Manager	Requests compliance audit trail for expense policy governance over 3 years.	Platform generates audit report: all policy versions, change dates, approvers, and notification records for 3-year period.	COMPLIANCE_REPORT_GENERATED

7.5 UC-BIZ-005: Organisational Change — Role Restructure as a Business Solution

UC-BIZ-005 — Organisational Restructure — Governed as a Business Solution

Domain: Business & Operational **Actors:** HR Director, Operations Director, Executive Sponsor, Affected Teams

Trigger: Organisation proposes a restructure of its technology operations function.

Outcome: Restructure designed, validated, and launched under governance. Post-implementation effectiveness measured against defined outcomes.

Step	Actor	Action	System Response	Event Published
1	HR Director	Creates solution record: type = Business Solution (Organisational Design). Attaches current state org chart (artefact), target state design, change rationale.	Solution record created. Org design versions tracked as artefacts.	SOLUTION_CREATED
2	HR Director	Raises CCR for the restructure. Attaches: impact assessment, redundancy risk assessment, legal compliance check (employment law).	CCR submitted. Risk = High. Routes to Executive Sponsor and Legal.	CCR_SUBMITTED
3	Executive Sponsor	Reviews restructure proposal. Approves.	CCR approved with executive rationale recorded.	CCR_APPROVED

4	HR Director	Executes restructure: team changes, role transitions, reporting line updates. Records completion milestones in platform.	Implementation milestones versioned. Change provenance maintained.	CHANGE_IMPLEMENTED
5	Platform	—	60 days post-implementation: AI Agent analyses available telemetry (engagement survey scores, productivity proxies) to assess restructure impact vs. intended outcomes.	AI_RECOMMENDATION_PUBLISHED (effectiveness_assessment)
6	HR Director	Reviews outcome assessment. Documents lessons learned as a linked artefact to the solution record.	Outcome artefact stored. Institutional learning preserved.	ARTEFACT_PUBLISHED (lessons_learned)

8 Research, Academic & Scientific Workflows

Use cases for research solutions: reproducible experiment pipelines, dataset versioning and governance, computational model development, publication workflows, and cross-institutional collaboration.

8.1 UC-RES-001: Reproducible Computational Experiment Pipeline

UC-RES-001 — Reproducible Computational Experiment — Environment to Attestation

Domain: Research & Academic **Actors:** Research Scientist, Principal Investigator, Platform

Trigger: A research scientist sets up a reproducible computational analysis pipeline for a clinical data study.

Outcome: Experiment executed in hermetic research SDE. Results attested with environment digest, dataset version, and script version. Independently reproducible from stored artefacts.

Step	Actor	Action	System Response	Event Published
1	Research Scientist	Creates solution record: type = Research Asset (Experimental Framework). Provisions Research SDE with specific Python environment, statistical libraries, and compute resources.	Research SDE provisioned from Nix/container definition. All software versions pinned and locked.	SOLUTION_CREATED → SDE_CREATED
2	Research Scientist	Registers dataset: creates Dataset Pointer artefact with external storage URL (institutional data repository), SHA-256 checksum, access credentials, dataset version.	Dataset pointer artefact stored. Checksum verification confirms dataset integrity at registration time.	ARTEFACT_PUBLISHED (dataset_pointer)
3	Research Scientist	Commits analysis scripts to SDE version control. Each script version has a commit hash.	Script versions tracked in SDE. Linked to the Research Asset solution record.	CONFIGURATION_VERSIONED
4	Research Scientist	Triggers experiment execution: platform runs scripts against registered dataset in hermetic research SDE.	Experiment pipeline executes. All dependencies loaded from pinned versions. No ambient state.	BUILD_STARTED (experiment_run)
5	Platform	—	Experiment completes. Result artefacts stored: output data, figures, statistical outputs,	ARTEFACT_PUBLISHED (results)

			execution log. SHA-256 digest for each result.	
6	Platform	—	Attestation generated: experiment input hash (dataset version + script version), environment definition hash, result hashes, execution timestamp, compute resource spec.	ATTESTATION_CREATED (experiment)
7	Principal Investigator	Reviews results. Requests independent replication by a second scientist.	Second scientist provisions identical Research SDE from same definition. Runs same pipeline. Results match within numerical tolerance. Reproducibility confirmed.	BUILD_PASSED (replication_confirmation)

8.2 UC-RES-002: Dataset Lifecycle — Curation, Versioning and Access Governance

UC-RES-002 — Research Dataset — Curation, Versioning and Governed Access

Domain: Research & Academic **Actors:** Data Curator, Research Scientist, Data Access Committee, IRB

Trigger: A clinical trial dataset must be curated, version-controlled, and made accessible to approved researchers.

Outcome: Dataset v1.0 curated, attested, and published under access governance. Approved researchers granted access with full audit trail of who accessed what and when.

Step	Actor	Action	System Response	Event Published
1	Data Curator	Creates solution record: type = Research Asset (Dataset). Registers dataset metadata: clinical trial ID, patient count, variables, collection period, data types.	Solution record created. Dataset pointer artefact created referencing secure storage.	SOLUTION_CREATED
2	Data Curator	Performs data curation: applies pseudonymisation, variable labelling, quality checks. Documents curation methodology.	Curation methodology artefact stored. Dataset quality report created.	ARTEFACT_PUBLISHED (curation_methodology, quality_report)
3	IRB	Reviews dataset for ethical compliance: confirms consent coverage, pseudonymisation	IRB approval artefact stored with approval date and reference number.	APPROVAL_RECORDED (IRB)

		adequacy, access control requirements.		
4	Data Curator	Sets access policy: approved research projects only, requires Data Access Committee (DAC) approval per access request.	Access policy version-controlled. Dataset lifecycle state → Available (Restricted Access).	SOLUTION_RELEASED
5	Research Scientist	Submits data access request: research project description, data variables needed, statistical purpose.	Access request routed to DAC.	CCR_SUBMITTED (data_access_request)
6	Data Access Committee	Reviews request. Approves with conditions: specific variables only, 12-month access, no re-identification attempts.	Access grant recorded with conditions. Researcher granted time-limited access.	CCR_APPROVED
7	Platform	—	All researcher access events logged in immutable audit trail: who accessed, which files, at what timestamps. Audit trail available for IRB review.	AUDIT_EVENTS_PUBLISHED

8.3 UC-RES-003: Computational Model — Development, Validation and Publication

UC-RES-003 — Computational Model — Peer-Reviewed Development and Publication

Domain: Research & Academic **Actors:** Research Scientist, Principal Investigator, Peer Reviewers, Journal Editor

Trigger: A research team develops a novel epidemiological simulation model and submits it for publication.

Outcome: Model v1.0 validated, attested, and published with full reproducibility record. Peer review integrated as a quality gate. All model inputs and outputs publicly accessible via DOI-linked artefacts.

Step	Actor	Action	System Response	Event Published
1	Research Scientist	Creates solution record: type = Research Asset (Computational Model). Provisions Research SDE. Model code versioned in SDE.	Solution record created. SDE provisioned. Initial model version registered.	SOLUTION_CREATED → SDE_CREATED

2	Research Scientist	Develops and iterates model. Each significant model version committed with: hyperparameters, training data version, validation results.	Model versions tracked. Validation artefacts stored per version.	ARTEFACT_PUBLISHED (per model version)
3	Research Scientist	Runs validation suite: cross-validation, sensitivity analysis, comparison vs. empirical data.	Validation results stored as artefacts. Model build attested.	ATTESTATION_CREATED → QUALITY_GATE_PASSED
4	Principal Investigator	Declares model ready for peer review. Submits to journal with attached model version, dataset version, and reproducibility package.	Submission artefact created. Release record opened with Peer Review gate.	RELEASE_PLANNED
5	Peer Reviewers	Two independent reviewers access reproducibility package. Run model in their own environment using locked definition.	Both reviewers confirm reproduction of key results. Peer review gate: PASS.	QUALITY_GATE_PASSED (peer_review)
6	Journal Editor	Accepts paper. Assigns DOI.	DOI registered as distribution record. Model state → Published. Artefacts publicly accessible.	DISTRIBUTION_EVENTS_PUBLISHED
7	Platform	—	Post-publication: any future model updates create new versions. Original published version immutably preserved and publicly accessible via DOI.	SOLUTION_VERSION_RELEASED (future updates)

8.4 UC-RES-004: Multi-Institutional Collaborative Research Pipeline

UC-RES-004 — Multi-Institutional Research Collaboration — Federated Data and Shared Pipeline

Domain: Research & Academic **Actors:** Principal Investigators (2 institutions), Data Stewards, Research Scientists

Trigger: Two universities collaborate on a joint longitudinal study requiring shared pipeline governance across institutional data repositories.

Outcome: Shared research pipeline operational. Each institution retains data sovereignty. Collaborative results attested and reproducible from both sites.

Step	Actor	Action	System Response	Event Published

1	PI (Institution A)	Creates shared Research Asset solution record with collaborative governance configuration. Invites Institution B as co-owner.	Solution record created with cross-institutional RBAC. Both institutions have governed access.	SOLUTION_CREATED (cross-institutional)
2	Data Stewards (both)	Each institution registers their local dataset as a Dataset Pointer artefact. Checksums verified. Access credentials scoped to the research project.	Two dataset pointers registered. Each institution's data remains on their infrastructure.	ARTEFACT_PUBLISHED (×2, dataset_pointers)
3	Research Scientists	Jointly develop analysis pipeline in shared SDE. All code changes require approval from at least one PI from each institution.	Pipeline code versioned. Approval requirement enforced by RBAC.	CONFIGURATION_VERSIONED
4	Platform	—	Pipeline executed at each site against local data. Results aggregated (not raw data). Aggregated result artefacts stored in shared registry.	BUILD_PASSED (×2, per site)
5	Research Scientists	Compare aggregated results. Merge into combined analysis. Validate statistical consistency across sites.	Combined analysis artefact stored. Validation results published.	ARTEFACT_PUBLISHED (combined_analysis)
6	PI (both)	Both PIs sign off on submission. Submit to journal with reproducibility package referencing both dataset pointers.	Joint sign-off recorded with identities from both institutions. Submission artefact created.	APPROVAL_RECORDED (joint)

8.5 UC-RES-005: Research Software — Open Source Release Governance

UC-RES-005 — Research Software — Governed Open Source Release

Domain: Research & Academic **Actors:** Research Scientist, Principal Investigator, Technology Transfer Office, Legal

Trigger: A research team wants to open-source a novel data analysis library developed during a funded research project.

Outcome: Library released as governed open source under Apache 2.0. Funding acknowledgement, licence, and IP clearance documented. All dependencies licence-compatible.

Step	Actor	Action	System Response	Event Published
1	Research Scientist	Raises CCR: type = Open Source Release. Attaches: library description, proposed licence (Apache 2.0), funding acknowledgements, list of dependencies.	CCR submitted. Routes to Technology Transfer Office and Legal.	CCR_SUBMITTED
2	Technology Transfer Office	Reviews IP ownership: confirms funded research IP policy allows open-source release under proposed terms.	IP clearance artefact stored.	APPROVAL_RECORDED (TTO)
3	Legal	Reviews all dependency licences for compatibility with Apache 2.0. Confirms SBOM is complete.	SBOM generated. All licences verified compatible. Legal approval recorded.	APPROVAL_RECORDED (Legal)
4	Research Scientist	Prepares release: adds LICENCE file, CITATION.cff, funding acknowledgements, contribution guidelines, API documentation.	Release artefacts prepared. SLSA attestation generated for source release.	ATTESTATION_CREATED
5	Principal Investigator	Final approval of open source release.	CCR approved. Release state → Approved.	CCR_APPROVED
6	Platform	—	Library published to open source registry (PyPI/CRAN/CRAN). DOI assigned. Publication record created in platform.	DISTRIBUTION_EVENTS_PUBLISHED
7	Platform	—	Library now managed as a Library solution type in Qala: version lifecycle governed, security scans automated, consumer notifications on updates.	SOLUTION_STATE_CHANGED → Active (open_source)

8.6 UC-RES-006: Retracted Research — Lifecycle Management

UC-RES-006 — Research Retraction — Governed Withdrawal and Downstream Impact

Domain: Research & Academic **Actors:** Principal Investigator, Journal Editor, Research Community, Platform

Trigger: A published research paper is found to have a critical methodological flaw requiring retraction.

Outcome: Retraction formally recorded. All downstream research that cited or built upon the retracted work identified and notified. Corrected analysis initiated.

Step	Actor	Action	System Response	Event Published
1	Principal Investigator	Submits retraction request to journal. Records in platform: retraction reason, affected paper DOI, nature of flaw.	CCR submitted: type = Publication Retraction. High severity.	CCR_SUBMITTED (retraction)
2	Journal Editor	Reviews retraction. Confirms and publishes retraction notice.	Retraction notice DOI attached. Journal approval recorded.	APPROVAL_RECORDED (journal)
3	Platform	—	Solution state changed to Retracted. Original published artefacts marked as Retracted (but preserved for audit).	SOLUTION_STATE_CHANGED → Retracted
4	Platform	—	Impact analysis: queries research relationship graph for solutions that declared a dependency on this research asset (datasets, pipelines, computational models built upon this work).	IMPACT_ANALYSIS_COMPLETE
5	Platform	—	Downstream notification: all identified researchers and solutions with dependencies on the retracted work receive notifications describing the retraction and recommending review.	CONSUMER_NOTIFICATIONS_SENT
6	Research Scientist	Initiates corrected analysis in new Research SDE. Corrected version treated as a new solution version with the flaw addressed.	New solution version created. Correction traceable to retraction record.	SOLUTION_CREATED (corrected_version)

9 Tools, Libraries, Packages & Toolchain Workflows

Use cases for tooling solutions: library publication governance, toolchain definition and approval, SDK release management, package deprecation, and SBOM generation.

9.1 UC-TOOL-001: Internal Library — Publish, Govern and Version

UC-TOOL-001 — Internal Library — Publish, Governance and Version Lifecycle

Domain: Tools & Libraries **Actors:** Library Author, Tech Lead, Security Engineer, Consumers

Trigger: An internal shared library is ready for its first governed publication to the internal package registry.

Outcome: Library v1.0.0 published with SBOM, SLSA attestation, licence declaration, and security scan results. All internal consumers can discover and adopt it.

Step	Actor	Action	System Response	Event Published
1	Library Author	Creates solution record: type = Library. Enters: language, licence, internal/external visibility, dependency list.	Solution record created. SBOM auto-generated from declared dependencies.	SOLUTION_CREATED
2	Library Author	Runs full CI pipeline: unit tests (98% coverage), SAST, SCA, dependency audit. All pass.	Pipeline gates pass. SLSA attestation generated.	BUILD_PASSED → ATTESTATION_CREATED
3	Security Engineer	Reviews SCA results and SAST findings. Confirms zero High/Critical issues.	Security sign-off recorded.	APPROVAL_RECORDED (Security)
4	Tech Lead	Reviews API design and documentation completeness.	Tech Lead approval recorded.	APPROVAL_RECORDED (Tech Lead)
5	Platform	—	Library v1.0.0 published to internal registry. SBOM attached to registry entry. Consumers can discover, inspect SBOM, and adopt with digest-pinned dependency reference.	DISTRIBUTION_EVENTS_PUBLISHED
6	Library Author	6 months later: publishes v1.1.0 (new feature, backward compatible). Releases v2.0.0 (breaking change) with consumer notification pipeline triggered.	All versions governed. Breaking change triggers UC-SW-005 consumer impact workflow.	CONSUMER_NOTIFICATIONS_SENT (v2.0.0)

9.2 UC-TOOL-002: Enterprise Toolchain — Definition, Approval and Enforcement

UC-TOOL-002 — Enterprise Toolchain — Definition, Approval and SDE Enforcement

Domain: Tools & Libraries **Actors:** Platform Engineer, Tech Lead Council, Security Engineer

Trigger: Platform Engineering defines the approved enterprise-wide toolchain for backend service development.

Outcome: Toolchain v1.0 approved. All new backend service SDEs automatically provisioned with approved toolchain. Deviation attempts blocked.

Step	Actor	Action	System Response	Event Published
1	Platform Engineer	Creates solution record: type = Toolchain (Backend Services). Defines toolchain: Go 1.22, golangci-lint 1.57, GoReleaser, Cosign, SLSA-github-generator, Prometheus client, OpenTelemetry SDK.	Solution record created. Each tool registered in Tool Registry with approved version and SHA-256 digest.	SOLUTION_CREATED
2	Tech Lead Council	Reviews toolchain composition. Votes to approve.	Toolchain CCR approved with council quorum.	CCR_APPROVED
3	Security Engineer	Validates all tools in toolchain against security policy: no known CVEs in approved versions, licence compatible, vendor security posture acceptable.	Security gate passed. All tool approvals recorded.	QUALITY_GATE_PASSED (Security)
4	Platform Engineer	Publishes toolchain as SDE template: all new backend service SDEs provisioned with this toolchain.	Toolchain v1.0 locked. SDE template updated.	SOLUTION_RELEASED
5	Developer	Attempts to install an unapproved tool in their backend service SDE.	Platform policy engine blocks install: "Tool X is not in the approved toolchain. Raise a Tool Addition CCR to request approval."	SECURITY_POLICY_VIOLATION_BLOCKED
6	Developer	Raises Tool Addition CCR requesting approval of a new linting tool.	CCR routed to Tech Lead Council. Approved after review. Tool added to toolchain v1.1.	CCR_APPROVED → SOLUTION_VERSION_RELEASED

9.3 UC-TOOL-003: SDK Release — Compatibility Matrix and Multi-Platform Distribution

UC-TOOL-003 — SDK Release — Compatibility Matrix Management and Distribution

Domain: Tools & Libraries **Actors:** SDK Author, QA Engineer, Release Manager, Developer Community

Trigger: A cross-platform SDK is releasing v3.0 with support for new platform targets and breaking API changes.

Outcome: SDK v3.0 released with compatibility matrix published. Breaking change consumers notified. Multi-platform artefacts distributed to all registries simultaneously.

Step	Actor	Action	System Response	Event Published
1	SDK Author	Creates release: SDK v3.0. Attaches: compatibility matrix (language × platform combinations supported), breaking changes list, migration guide.	Release record created. Compatibility matrix stored as artefact.	RELEASE_PLANNED
2	QA Engineer	Runs matrix test suite: executes SDK test suite against every language × platform combination in the compatibility matrix (12 combinations).	Matrix test results stored. All 12 combinations pass.	QUALITY_GATE_PASSED (matrix)
3	Platform	—	Consumer impact analysis: identifies all solutions consuming SDK v2.x. Breaking change consumers notified.	CONSUMER_NOTIFICATIONS_SENT
4	Release Manager	Approves release.	Release state → Approved.	RELEASE_APPROVED
5	Platform	—	Simultaneous distribution: SDK v3.0 artefacts (per platform/language) published to: npm, PyPI, Maven Central, NuGet, crates.io, Swift Package Index. All with matching SLSA attestations.	DISTRIBUTION_EVENTS_PUBLISHED (×6 registries)
6	Platform	—	SDK v2.x enters deprecation window (90 days). Deprecation header added to all v2.x responses.	SOLUTION_DEPRECATED

9.4 UC-TOOL-004: Package Security Incident — CVE Response Across All Consumers

UC-TOOL-004 — Package Security Incident — Zero-Day CVE Emergency Response

Domain: Tools & Libraries **Actors:** SEM, Security Engineer, Platform, All Consumer Developers

Trigger: A critical zero-day CVE is published affecting a widely used internal library that is consumed by 47 solutions.

Outcome: All 47 consumer solutions identified, prioritised, and patched within 24-hour SLA. Patched library version attested and redistributed.

Step	Actor	Action	System Response	Event Published
1	SEM	—	NVD publishes CVE with CVSS 9.8 (Critical). SEM detects match in library artefact registry.	VULNERABILITY_FOUND → SECURITY_ALERT_SENT
2	Platform	—	Consumer impact blast radius: queries solution relationship graph. Returns 47 solutions consuming affected version. Grouped by severity exposure: 12 production, 28 staging, 7 development.	IMPACT_ANALYSIS_COMPLETE
3	Security Engineer	Confirms severity. Declares security incident. Sets SLA: 24h for all affected consumers.	Emergency mode activated. All 47 consumer teams and Tech Leads notified.	SECURITY_ALERT_SENT (priority=critical)
4	Library Author	Patches library. Fast-track pipeline: build, SAST, SCA (confirms CVE resolved), SLSA attestation. Publishes patched version.	Patched library v1.4.1 published in < 4 hours.	BUILD_PASSED → ATTESTATION_CREATED → DISTRIBUTION_EVENTS_PUBLISHED
5	Developers (×47)	Each consumer team updates dependency to v1.4.1. Rebuilds. Pipeline confirms CVE resolved.	Each rebuild generates new SLSA attestation with patched dependency.	BUILD_PASSED (×47)
6	Security Engineer	Confirms all 47 consumer solutions resolved within 24-hour SLA.	Incident closed. Resolution timeline recorded. All 47 solutions show CVE status = Resolved.	INCIDENT_CLOSED

9.5 UC-TOOL-005: Reference Architecture — Publish, Adopt and Govern Updates

UC-TOOL-005 — Reference Architecture — Governed Publication and Consumer Adoption

Domain: Tools & Libraries **Actors:** Enterprise Architect, Tech Lead Council, Consuming Teams

Trigger: Enterprise Architecture publishes a new microservices reference architecture for adoption across all product teams.

Outcome: Reference architecture v1.0 published. Consuming teams register adoption. Updates governed by CCR with consumer impact assessment.

Step	Actor	Action	System Response	Event Published
1	Enterprise Architect	Creates solution record: type = Resource (Reference Architecture). Publishes: architecture diagrams, ADRs, technology choices, sample code, runbook templates.	Solution record created. All components stored as versioned artefacts.	SOLUTION_CREATED
2	Tech Lead Council	Reviews and approves reference architecture.	Council approval recorded.	APPROVAL_RECORDED
3	Platform	—	Reference architecture v1.0 published to internal resource catalogue. All product teams notified.	SOLUTION_RELEASED
4	Product Teams	Teams declare adoption: each team creates a solution relationship (their service → this reference architecture).	Adoption relationships recorded. 18 teams adopt within 30 days.	RELATIONSHIPS_CREATED (×18)
5	Enterprise Architect	12 months later: raises CCR to update reference architecture for new cloud-native security requirements.	Impact analysis: 18 adopting teams identified. All notified of incoming change.	CCR_SUBMITTED → IMPACT_ANALYSIS_COMPLETE
6	Adopting Teams	Each team reviews impact on their specific solution. 16 of 18 can adopt immediately; 2 need migration assistance.	Migration support tracked per team. CCR approved after all teams confirm readiness plan.	CCR_APPROVED → SOLUTION_VERSION_RELEASED

10 Legal, Tax & Regulatory Solution Workflows

Use cases for legal, tax, and regulatory solutions: legislative change governance, jurisdiction variant management, contract template lifecycle, regulatory product updates, and compliance framework management.

10.1 UC-LEG-001: Contract Template — Version Control and Jurisdiction Governance

UC-LEG-001 — Contract Template — Version Control and Jurisdiction Variant Management

Domain: Legal, Tax & Regulatory **Actors:** Legal Operations Manager, Commercial Lawyer, Compliance Officer

Trigger: Legal team manages a master NDA template that has jurisdiction-specific variants across 12 markets.

Outcome: NDA template and all 12 jurisdiction variants version-controlled. Any change to master propagates to variants through governed review. All versions auditable.

Step	Actor	Action	System Response	Event Published
1	Legal Operations Manager	Creates solution record: type = Legal Solution (Contract Template). Registers master NDA template and 12 jurisdiction variant relationships.	Solution record created. Master template and 12 variants stored as versioned artefacts.	SOLUTION_CREATED
2	Commercial Lawyer	Proposes update to master NDA: updated limitation of liability clause.	CCR submitted: type = Contract Amendment. Impact analysis: identifies which of the 12 jurisdiction variants inherit this clause.	CCR_SUBMITTED → IMPACT_ANALYSIS_COMPLETE
3	Commercial Lawyer	Reviews impact analysis. 8 variants need updating (inherit clause). 4 variants have local law overrides (not affected).	CCR scope confirmed: 8 variants to be updated alongside master.	CCR_APPROVED
4	Lawyers (jurisdiction-specific)	Each jurisdiction lawyer reviews proposed clause update for local law compliance. All 8 approve.	8 jurisdiction approvals recorded.	APPROVAL_RECORDED (×8)
5	Platform	—	Master template v2.0 published. 8 variant v2.0s published simultaneously. 4 unaffected variants unchanged.	SOLUTION_VERSION_RELEASED (master + 8 variants)

6	Legal Operations Manager	Previous contract versions remain accessible for contracts already executed under v1.0.	Version immutability ensures any dispute about contract terms can be audited against the version active at signing.	AUDIT_TRAIL_COMPLETE

10.2 UC-LEG-002: Regulatory Product — Change Triggered by External Legislation

UC-LEG-002 — Regulatory Product — Externally-Triggered Legislative Change

Domain: Legal, Tax & Regulatory **Actors:** Regulatory Product Manager, Legal, Compliance Officer, Regulatory Intelligence Service

Trigger: New data protection legislation is enacted with a 90-day compliance window.

Outcome: All affected products identified. Required changes implemented and released before compliance deadline. Zero missed obligations.

Step	Actor	Action	System Response	Event Published
1	Regulatory Intelligence Service	New legislation enacted: Regional AI Governance Act, effective in 90 days.	Integration (or manual creation): Legislative Change CCR auto-generated referencing the legislation. Affected solution types flagged: Platform, Application, AI-powered Service solutions.	CCR_SUBMITTED (legislative)
2	Compliance Officer	Reviews legislation. Identifies specific obligations: AI system registration, conformity assessment, transparency documentation.	Obligation list created as structured artefact. Each obligation mapped to the specific affected solutions.	ARTEFACT_PUBLISHED (obligation_map)
3	Regulatory Product Manager	Assigns implementation tasks per obligation per affected solution. Sets completion deadlines aligned with compliance window.	Implementation plan version-controlled. Progress tracked in platform.	CONFIGURATION_VERSIONED (implementation_plan)
4	Development Teams	Implement required changes per obligation for each affected solution. Each change raised as a CCR tied to the legislative obligation.	Implementation CCRs approved and implemented. Changes traceable to legislative obligation.	CCR_APPROVED → CHANGE_IMPLEMENTED (per obligation)
5	Compliance Officer	Verifies all obligations implemented before deadline. Generates	Compliance evidence report generated: each obligation mapped to implementation	COMPLIANCE_REPORT_GENERATED

		compliance evidence report.	CCR, with approval records and implementation dates.	
6	Platform	—	Compliance deadline reached. All obligations implemented and evidenced. Regulatory risk status = Compliant.	COMPLIANCE_EVENTS_PUBLISHED

10.3 UC-LEG-003: Tax Product — Multi-Jurisdiction Rule Management

UC-LEG-003 — Tax Product — Multi-Jurisdiction Rule Management and Variant Release

Domain: Legal, Tax & Regulatory **Actors:** Tax Product Manager, Tax Technologist, Legal, Compliance Officer

Trigger: Annual tax year change requiring updates across 15 jurisdiction variants of a corporate tax product.

Outcome: All 15 jurisdiction variants updated, validated, and released with effective-date scheduling. Zero jurisdiction misses.

Step	Actor	Action	System Response	Event Published
1	Tax Product Manager	Creates master CCR: Annual Tax Year Update. Attaches: update summary per jurisdiction (15 jurisdictions, sourced from regulatory monitoring feeds).	Master CCR created. 15 jurisdiction-specific sub-CCRs auto-generated from master.	CCR_SUBMITTED (master + 15 sub-CCRs)
2	Tax Technologist (per jurisdiction)	Updates rule set for each jurisdiction in Tax Solution SDE: new rates, changed thresholds, new relief provisions.	15 new jurisdiction rule set versions created in parallel.	CONFIGURATION_VERSIONED (×15)
3	Tax Technologist	Runs validation suite per jurisdiction: correctness tests (test cases per jurisdiction rule), regression tests, cross-jurisdiction consistency checks.	All 15 jurisdictions pass validation. Quality gate passed.	QUALITY_GATE_PASSED (×15)
4	Legal (per jurisdiction)	Jurisdiction-specific legal review for the 5 jurisdictions with complex legislative changes.	5 jurisdiction legal approvals recorded. 10 jurisdiction updates approved at Tax Product Manager level.	APPROVAL_RECORDED (×15 total)
5	Tax Product Manager	Sets effective dates per jurisdiction (each has different tax year start date). Creates 15 effective-date releases.	15 release records created with jurisdiction-specific activation timestamps.	RELEASE_APPROVED (×15, pending effective dates)

6	Platform	—	Releases activate on their respective effective dates. Each jurisdiction's rule set activates at local midnight on the tax year start. All consumers updated simultaneously per jurisdiction.	DEPLOYED (×15, staggered by effective date)

10.4 UC-LEG-004: Compliance Framework — Internal Policy Lifecycle

UC-LEG-004 — Internal Compliance Framework — Policy Development and Enforcement

Domain: Legal, Tax & Regulatory **Actors:** Chief Compliance Officer, Policy Author, Compliance Team, Impacted Business Units

Trigger: Organisation must develop and publish an updated Information Security Policy to align with ISO 27001:2022.

Outcome: Information Security Policy v3.0 published. All business units confirmed for acknowledgement. Compliance evidence of policy coverage generated.

Step	Actor	Action	System Response	Event Published
1	Policy Author	Creates solution record: type = Compliance Framework (Information Security Policy). Provisions Knowledge SDE for document authoring.	Solution record created.	SOLUTION_CREATED
2	Policy Author	Authors policy update addressing all new ISO 27001:2022 controls. Each section version-controlled.	Policy document versioned. Control coverage mapping artefact created.	ARTEFACT_PUBLISHED
3	Chief Compliance Officer	Reviews policy. Approves with minor edits.	CCO approval recorded.	APPROVAL_RECORDED (CCO)
4	Legal	Reviews policy for employment law and regulatory obligations alignment.	Legal approval recorded.	APPROVAL_RECORDED (Legal)
5	Platform	—	Policy v3.0 published. All business units and employees notified. Acknowledgement tracking activated.	DISTRIBUTION_EVENTS_PUBLISHED
6	Compliance Team	Monitors acknowledgement completion rate. After 30 days: 96%	Acknowledgement completion recorded as quality event.	QUALITY_EVENTS_PUBLISHED

		acknowledged; 4% pending.		
7	Platform	—	ISO 27001 compliance gate for this control: policy coverage = PASS (all controls addressed in policy). Evidence record generated for annual audit.	COMPLIANCE_GATE_PASSED

10.5 UC-LEG-005: Legal Hold — Evidence Preservation for Litigation

UC-LEG-005 — Legal Hold — Evidence Preservation and Access Control for Litigation

Domain: Legal, Tax & Regulatory **Actors:** General Counsel, Legal Operations, Platform Admin

Trigger: Organisation receives a litigation hold notice requiring preservation of all relevant records.

Outcome: Legal hold applied. All relevant artefacts, records, and audit trails frozen from deletion. Privilege-protected materials segregated. Legal team has governed access.

Step	Actor	Action	System Response	Event Published
1	General Counsel	Creates Legal Hold record in platform: case reference, relevant date range, relevant solution types (applications, business solutions, communications data), custodians.	Legal hold record created. Preservation scope defined.	LEGAL_HOLD_CREATED
2	Platform	—	Preservation applied: all artefacts, audit records, configuration versions, CCRs, quality records, and distribution records within scope are locked from deletion or modification.	RECORDS_PRESERVED
3	Platform	—	Automatic retention extension: any records within scope that would have expired under normal retention policy are extended until legal hold is released.	RETENTION_EXTENDED
4	Legal Operations	Identifies privilege-protected communications (attorney-client). Applies privilege flag — segregated from normal discovery scope.	Privilege designation recorded per artefact. Flagged records excluded from standard discovery exports.	PRIVILEGE_FLAGS_APPLIED

5	General Counsel	Generates discovery production: exports all in-scope non-privileged records in agreed format for opposing counsel.	Export generated with complete provenance metadata. Export event audited.	AUDIT_EVENTS_PUBLISHED (export)
6	General Counsel	Litigation resolved. Releases legal hold.	Legal hold released. Normal retention policies resume. Hold record preserved for 7 years.	LEGAL_HOLD_RELEASED

11 Platform & Ecosystem Workflows

Use cases for platform solutions: marketplace launch and governance, ecosystem onboarding, multi-sided platform partner management, and developer ecosystem release cycles.

11.1 UC-PLT-001: Two-Sided Marketplace — Governed Launch

UC-PLT-001 — Two-Sided Marketplace — Platform Launch and Partner Onboarding

Domain: Platform & Ecosystem **Actors:** Platform Manager, Legal, Compliance Officer, Supply-Side Partners, Demand-Side Users

Trigger: A new B2B agricultural marketplace connecting buyers and sellers is ready to launch.

Outcome: Marketplace platform governed through launch. Supply-side partners onboarded. Demand-side users registered. Platform quality gates passed. Launch event published.

Step	Actor	Action	System Response	Event Published
1	Platform Manager	Creates solution record: type = Platform (Marketplace). Defines: platform scope, transaction model, partner onboarding criteria, dispute resolution policy.	Solution record created. Platform policies stored as versioned artefacts.	SOLUTION_CREATED
2	Platform Manager	Configures supply-side partner onboarding workflow: verification, contract execution, catalogue publication approval.	Onboarding workflow versioned. Partner template contract published.	CONFIGURATION_VERSIONED
3	Legal	Reviews marketplace terms: buyer terms, seller terms, platform liability policy.	Legal approval recorded.	APPROVAL_RECORDED
4	Compliance Officer	Reviews platform for compliance: AML for transaction flows, data protection (GDPR), agricultural regulatory requirements for listed products.	Compliance gate passed. Regulatory assessment stored as artefact.	COMPLIANCE_GATE_PASSED
5	Platform Manager	Onboards 15 founding supply-side partners. Each partner undergoes verification workflow.	15 partner onboarding records created. Each partner KYC status recorded.	DISTRIBUTION_EVENTS_PUBLISHED (partner_onboarding)
6	Platform Manager	Initiates platform launch. Release gates passed. Platform state → Live.	Platform released. Launch event published.	SOLUTION_RELEASED

7	Platform	—	Platform telemetry active: transaction volume, partner engagement, buyer acquisition, dispute rate. AI Agent begins anomaly detection and optimisation recommendations.	AI_MONITORING_ACTIVE

11.2 UC-PLT-002: Developer Ecosystem — API Platform Governance

UC-PLT-002 — Developer Ecosystem — API Platform Lifecycle Governance

Domain: Platform & Ecosystem **Actors:** Platform Architect, Developer Relations, External Developer Community

Trigger: An API platform managing third-party developer integrations needs to deprecate an old API version while launching a new one.

Outcome: New API version launched. Developer community notified. Migration window set. Old version traffic monitoring shows migration complete before deprecation deadline.

Step	Actor	Action	System Response	Event Published
1	Platform Architect	Creates solution record for API v3.0 with release of new platform capabilities: expanded query interface, webhook support, improved rate limits.	New API version registered. Release record created.	RELEASE_PLANNED
2	Developer Relations	Prepares: migration guide, code samples in 5 languages, sandbox environment for testing new API, FAQ for breaking change questions.	Migration resource artefacts stored and linked to release record.	ARTEFACT_PUBLISHED (migration_resources)
3	Platform Architect	Approves launch of v3.0.	Release approved.	RELEASE_APPROVED
4	Platform	—	v3.0 deployed. Developer portal updated. All registered developers notified of new version with migration guide and 180-day deprecation window for v2.0.	DEPLOYED → CONSUMER_NOTIFICATIONS_SENT
5	Platform	—	Weekly migration progress monitoring: tracks v2.0 vs. v3.0 API call ratios. AI Agent flags developers with high v2.0 dependency who have not yet migrated.	AI_RECOMMENDATION_PUBLISHED (migration_risk_developers)

6	Developer Relations	Proactively contacts at-risk developers with migration support offer. 97% migrate before deadline.	Migration completions recorded.	DISTRIBUTION_EVENTS_PUBLISHED (migrations)
7	Platform	—	At deprecation deadline: v2.0 retired. 3 remaining v2.0 callers automatically redirected with 301 and deprecation notice.	SOLUTION_RETIRED

11.3 UC-PLT-003: Ecosystem — Third-Party Integration Certification

UC-PLT-003 — Platform Ecosystem — Third-Party Integration Certification

Domain: Platform & Ecosystem **Actors:** Integration Partner, Platform Certification Team, Security Engineer

Trigger: A third-party software vendor wants to certify their CRM integration against the Qala API.

Outcome: Integration certified. Partner listed in certified partner directory. Integration artefacts signed and published. Recertification schedule defined.

Step	Actor	Action	System Response	Event Published
1	Integration Partner	Creates partner solution record: type = Platform (Integration). Submits certification application with: integration specification, security documentation, test environment access.	Partner record created. Certification workflow initiated.	SOLUTION_CREATED (partner_integration)
2	Platform Certification Team	Reviews integration specification: API usage patterns, authentication model, data handling, error handling, retry behaviour.	Specification review artefact stored.	ARTEFACT_PUBLISHED (spec_review)
3	Security Engineer	Conducts security review: penetration test of integration endpoints, credential handling, data at rest and in transit.	Security test artefact stored. No critical findings.	QUALITY_GATE_PASSED (Security)
4	Platform Certification Team	Runs integration test suite against partner's test environment: functional correctness, API contract compliance,	Integration test results stored. All tests pass.	QUALITY_GATE_PASSED (Integration)

		performance under load.		
5	Platform Certification Team	Certifies integration. Signs certification artefact.	Certification artefact signed and stored. Partner integration listed in certified directory.	SOLUTION_RELEASED (certified)
6	Platform	—	Recertification scheduled: annual review triggered by calendar event or any breaking API change. Certification expiry date visible in partner directory.	CONFIGURATION_VERSIONED (recertification_schedule)

11.4 UC-PLT-004: Multi-Tenant SaaS — Tenant Onboarding and Isolation Governance

UC-PLT-004 — Multi-Tenant SaaS — Tenant Onboarding and Isolation Configuration

Domain: Platform & Ecosystem **Actors:** Platform Admin, New Tenant (Enterprise Customer), Security Engineer

Trigger: A new enterprise customer signs up for Qala at the Enterprise isolation tier.

Outcome: Tenant onboarded with dedicated cluster isolation, data residency configured, custom RBAC configured, and SSO integrated. All configuration version-controlled.

Step	Actor	Action	System Response	Event Published
1	Platform Admin	Creates new tenant record: isolation tier = Enterprise (dedicated cluster), data residency = EU (Frankfurt region), custom subdomain, SSO configuration.	Tenant provisioning workflow initiated. Dedicated cluster provisioned in specified region.	TENANT_CREATED
2	Platform	—	Dedicated cluster provisioned: isolated Kubernetes namespace, isolated database, isolated event bus partitions, encryption keys provisioned in customer-managed KMS.	TENANT_INFRASTRUCTURE_PROVISIONED
3	Platform Admin	Configures tenant RBAC: imports customer role definitions, maps to Qala standard roles, configures approval chains for enterprise governance.	RBAC configuration version-controlled.	CONFIGURATION_VERSIONED
4	Security Engineer	Validates tenant isolation: penetration	Isolation validation artefact stored. Security gate passed.	QUALITY_GATE_PASSED (Isolation)

		test confirming cross-tenant data inaccessibility. Audit trail isolation confirmed.		
5	Platform Admin	Configures SSO integration: connects customer's IdP (Okta/Azure AD) via SAML/OIDC.	SSO integration tested and validated. All user logins via customer IdP.	CONFIGURATION_VERSIONED (SSO)
6	New Tenant	Enterprise admin accesses platform. Creates first Solution Factory. Begins onboarding their team.	Tenant operational. First Solution Factory created.	SOLUTION_FACTORY_CREATED

12 Cross-Domain & Universal Workflows

Use cases demonstrating workflows that span multiple solution types and domains — the workflows that are unique to a universal solution management platform and impossible to execute coherently in any single-domain tool.

12.1 UC-XD-001: Universal Compliance Evidence Generation

UC-XD-001 — Universal Compliance Evidence — Multi-Framework, Multi-Domain Audit Package

Domain: Cross-Domain **Actors:** Chief Compliance Officer, Platform (automated), External Auditor

Trigger: A diversified enterprise operating across software, CPG, and financial services needs a single SOC 2 compliance package covering all solution domains.

Outcome: Single unified compliance evidence package generated covering all three business domains from one platform. Audit preparation time reduced from 6 weeks to 4 hours.

Step	Actor	Action	System Response	Event Published
1	Chief Compliance Officer	Initiates compliance package generation: scope = All solution factories, framework = SOC 2 Type II, period = previous 12 months.	Platform queues evidence collection across all enrolled Solution Factories and all solution types.	AUDIT_INITIATED
2	Platform	—	Evidence collection — Software: CI/CD pipeline gate results, deployment records, change management CCRs, security scan results, vulnerability management records, SDE access logs.	COMPLIANCE_EVIDENCE_GATHERED (software_domain)
3	Platform	—	Evidence collection — CPG: formulation change CCRs, batch quality records, GMP policy compliance checks, recall readiness test results, supplier certification records.	COMPLIANCE_EVIDENCE_GATHERED (cpg_domain)
4	Platform	—	Evidence collection — Financial: product governance CCRs, model validation records, MiFID II compliance gate results, audit trail completeness checks.	COMPLIANCE_EVIDENCE_GATHERED (financial_domain)
5	Platform	—	Unified SOC 2 evidence package generated: each SOC 2 control mapped to specific evidence records from all three domains. Controls with gaps identified and flagged.	COMPLIANCE_REPORT_GENERATED

6	Chief Compliance Officer	Reviews report. Adds manual evidence for 2 controls that require physical inspection artefacts (data centre visit record, HR background check policy).	Manual evidence attached. Package complete.	ARTEFACT_PUBLISHED (manual_supplements)
7	External Auditor	Receives package. Conducts audit. Raises 2 minor observations (no significant findings). SOC 2 Type II report issued.	Audit completion recorded. SOC 2 report stored as artefact linked to the compliance package.	AUDIT_COMPLETED

12.2 UC-XD-002: AI Agent — Cross-Domain Quality Pattern Learning

UC-XD-002 — AI Cross-Domain Quality Learning — Pattern Transfer Across Solution Types

Domain: Cross-Domain **Actors:** AI Agent, Data Scientists, Solution Owners across all domains

Trigger: AI Agent identifies a quality improvement pattern in software solutions that is applicable to CPG product development.

Outcome: Cross-domain quality insight surfaced. CPG team adopts adapted testing practice. Quality improvement measured.

Step	Actor	Action	System Response	Event Published
1	AI Agent	—	Analyses quality patterns across all solution types. Detects: software teams that run property-based testing (randomised input generation) have 34% fewer edge-case defects than those that do not.	AI_PATTERN_DETECTED (cross-domain)
2	AI Agent	—	Identifies structural analogous problem in CPG domain: formulation teams that test only defined test cases miss edge-case parameter combinations. Generates cross-domain insight recommendation.	AI_RECOMMENDATION_PUBLISHED (cross_domain)
3	AI Agent	—	Recommendation surfaces to CPG quality managers: "Formulation testing patterns from software quality engineering suggest randomised parameter space exploration (analogous to property-based testing) could identify stability edge cases	AI_RECOMMENDATION_PUBLISHED (cpg_specific)

			missed by current fixed test plan."	
4	QA Manager (CPG)	Reviews recommendation. Finds compelling. Raises CCR to pilot randomised parameter testing on 3 formulations.	CCR approved. Pilot QA methodology change recorded.	CCR_APPROVED
5	QA Team (CPG)	Runs pilot. Discovers 2 stability issues at parameter combination edges that were not caught by standard test plan.	2 defects created and addressed. Pilot outcome recorded.	DEFECT_CREATED → RESOLVED
6	QA Manager (CPG)	Approves adoption of adapted testing methodology. Raises CCR to update QA Standard Operating Procedure.	Updated QA SOP version published. Quality improvement attributed to cross-domain AI learning.	SOLUTION_VERSION_RELEASED (QA_SOP)

12.3 UC-XD-003: Supply Chain Platform — Multi-Solution-Type Integration

UC-XD-003 — Supply Chain Platform — Governing a Multi-Type Solution Ecosystem

Domain: Cross-Domain **Actors:** Supply Chain Platform Manager, Suppliers (CPG), Logistics Partners (Service), Financial Settlement (Financial Instrument)

Trigger: A supply chain platform integrates three solution types in one governed workflow: physical goods (product shipments), services (logistics), and financial instruments (payment terms).

Outcome: All three solution types governed, versioned, and linked. Complete supply chain event from product batch release through logistics service delivery to financial settlement fully traceable.

Step	Actor	Action	System Response	Event Published
1	CPG Supplier	Batch of physical goods released (UC-CPG-003 executed). Distribution event published with batch ID and retail destination.	DISTRIBUTION_EVENTS_PUBLISHED (physical_goods)	
2	Logistics Partner	Logistics service picks up shipment. Logistics Service solution record creates a service engagement linked to the physical goods distribution event.	Service engagement record created and linked to goods distribution event. Cold chain monitoring begins.	SOLUTION_CREATED (logistics_engagement)

3	Platform	—	IoT cold chain sensors in transit publish temperature events to platform event bus. Any breach of temperature threshold triggers an alert and potential quality hold.	DOMAIN_EVENTS_PUBLISHED (cold_chain_telemetry)
4	Logistics Partner	Delivery completed. Proof of delivery recorded. Logistics service engagement state → Delivered.	Service delivery event published.	DISTRIBUTION_EVENTS_PUBLISHED (logistics_delivery)
5	Financial Platform	On delivery confirmation: payment terms trigger. Financial instrument (trade finance instrument) activates settlement process.	Financial instrument solution event published. Settlement calculation initiated.	DOMAIN_EVENTS_PUBLISHED (financial_settlement)
6	Platform	—	Complete cross-domain provenance chain: physical goods batch → logistics service delivery → financial settlement. Any audit query can traverse the full chain across all three solution types from a single platform query.	AUDIT_TRAIL_COMPLETE (cross-domain)

12.4 UC-XD-004: Platform-Wide Security Incident Response

UC-XD-004 — Platform-Wide Security Incident — Cross-Domain Triage and Response

Domain: Cross-Domain **Actors:** SEM, CISO, Security Engineers, All Domain Teams

Trigger: SEM detects a credential compromise affecting a developer who has access across software, financial, and research solution factories.

Outcome: Compromised access revoked across all domains within 5 minutes. All sessions invalidated. Forensic review across all affected solution types completed. Incident closed with full audit trail.

Step	Actor	Action	System Response	Event Published
1	SEM	—	Detects anomalous access pattern: same user account accessing software SDEs (normal) and financial research SDEs (unusual, off-hours) and research datasets (unusual, bulk download attempt).	THREAT_DETECTED (cross-domain_anomaly)
2	SEM	—	All active sessions for compromised identity immediately invalidated	SDE_QUARANTINED (all affected SDEs)

			across ALL Solution Factories and ALL solution types within 90 seconds.	
3	Platform	—	CISO and domain security leads notified. Scope: 3 software SDEs quarantined, 1 financial research SDE quarantined, 2 research dataset access sessions terminated.	SECURITY_ALERT_SENT (priority=critical)
4	Security Engineer	Reviews access logs across all affected domains: software pipeline access (legitimate), financial research SDE (accessed unusual artefacts), research dataset (bulk download attempt, incomplete).	Cross-domain forensic report created. Blast radius confirmed: no data exfiltration confirmed for software or financial domains; potential partial research dataset access.	ARTEFACT_PUBLISHED (forensic_report)
5	CISO	Notifies research dataset data custodian. IRB notified per data governance protocol. Password reset forced for compromised identity.	Notifications recorded with timestamps. Password reset completion confirmed.	SECURITY_EVENTS_PUBLISHED
6	Security Engineer	Provisions clean replacement SDEs for the affected user from last known-good snapshots.	Clean SDEs provisioned. User access restored under re-authenticated session.	SDE_CREATED (clean_replacements)
7	CISO	Reviews incident. Determines phishing was likely vector. Triggers phishing awareness campaign for all users.	Campaign CCR created. Incident closed with lessons learned recorded.	INCIDENT_CLOSED

12.5 UC-XD-005: Universal Onboarding — New Solution Across Multiple Types

UC-XD-005 — New Organisation Onboarding — Multi-Domain Solution Factory Setup

Domain: Cross-Domain **Actors:** Platform Admin, CTO, Domain Leads (Software, CPG, Service)

Trigger: A diversified company onboards Qala to manage all their solution types: software products, physical goods, and managed services.

Outcome: Three Solution Factories configured across three domains. Initial SDEs provisioned. First solutions registered. Platform operational across all domains within one business day.

Step	Actor	Action	System Response	Event Published
1	Platform Admin	Provisions tenant. Creates three Solution Factories: Software Products, Physical Goods, and Managed Services.	Tenant created. 3 Solution Factories created with appropriate governance policies per domain.	TENANT_CREATED → SOLUTION_FACTORY_CREATED (×3)
2	CTO	Invites domain leads. Software Lead: Platform Engineer role. CPG Lead: Formulation Chemist role. Service Lead: Service Architect role.	Domain-specific roles assigned. Each lead sees portal customised to their domain vocabulary and relevant workflows.	CONFIGURATION_VERSIONED (RBAC)
3	Platform Engineer	Provisions first software SDE from approved template. Creates first Application solution record.	Software SDE live. First application registered.	SDE_CREATED → SOLUTION_CREATED (software)
4	Formulation Chemist	Provisions first Lab SDE. Creates first Consumer Packaged Good solution record. Begins importing legacy formulation specifications.	CPG SDE live. First formulation registered. Legacy specs imported as initial artefact versions.	SDE_CREATED → SOLUTION_CREATED (cpg)
5	Service Architect	Creates first Managed Service solution record. Publishes service specification to internal catalogue.	Service solution created. Service catalogue entry published.	SOLUTION_CREATED (service) → SOLUTION_RELEASED
6	Platform	—	AI Agent activated across all three domains. Baseline learning begins. Platform recommends first optimisations for each domain within 24 hours.	AI_MONITORING_ACTIVE (cross-domain)

12.6 UC-XD-006: Governed Retirement — End-of-Life Across Solution Types

UC-XD-006 — Governed Solution Retirement — End-of-Life Workflow for Any Solution Type

Domain: Cross-Domain **Actors:** Solution Owner, Release Manager, Consumer Organisations, Platform Admin

Trigger: An organisation retires three solutions simultaneously: a software API, a CPG product line, and a consulting methodology.

Outcome: All three solutions retired through governed workflows appropriate to their type. Consumer migration windows completed. All artefacts archived. Audit trail of retirement decisions preserved.

Step	Actor	Action	System Response	Event Published
1	Solution Owners (×3)	Each owner raises a Retirement CCR for their solution: Software API v2 (superseded by v3), CPG Product Line (market exit), Consulting Methodology (replaced by updated version).	3 Retirement CCRs submitted. Each routed per type-specific approval chain.	CCR_SUBMITTED (×3, retirement)
2	Platform	—	Consumer impact analysis for each: API v2 has 12 registered consumers; CPG Product Line has 4 retail distribution partners; Methodology has 8 active engagements referencing it.	IMPACT_ANALYSIS_COMPLETE (×3)
3	Release Manager (API)	Sets API v2 deprecation: 90-day notice period. Affected API consumers notified with migration guide to v3.	CONSUMER_NOTIFICATIONS_SENT (API consumers)	
4	CPG Ops Manager	Initiates product line market withdrawal: retail partners notified with sell-through period. Final batch production completed. Distribution suspended for new orders.	DISTRIBUTION_SUSPENDED (CPG product line)	
5	Knowledge Manager	Methodology retirement: active engagement references migrated to replacement methodology. Old methodology marked as superseded in knowledge base.	SOLUTION_DEPRECATED (methodology)	
6	Platform	—	At end of each retirement window: solutions transitioned to Retired state. Artefacts archived to cold storage (but fully queryable). Retirement CCRs and decisions preserved in immutable audit trail.	SOLUTION_STATE_CHANGED → Retired (×3)
7	Platform	—	All consumer notifications confirmed received. All migration windows complete. Retirement events published.	AUDIT_TRAIL_COMPLETE (retirements)

13 Universal Workflow Performance Metrics

Key performance targets for the most critical workflows across all solution domains.

Workflow Category	Key Metric	Industry Baseline	Qala Target
SDE Provisioning (all types)	Time to active SDE from approval	2–14 days (manual)	< 10 min (software); < 30 min (other types)
Hermetic Build Cycle	Commit to attested artefact	30–90 minutes	< 15 minutes
Change Control (standard)	CCR submission to approval	1–5 days	< 4 hours (standard); same day (most)
Change Control (regulated, high-risk)	CCR submission to multi-stage approval	1–6 weeks manual	< 5 days with governed workflow
Software Release (blue/green)	Release approval to live traffic	1–4 hours	< 30 minutes
Software Rollback	Incident detection to rollback complete	30–240 minutes	< 5 minutes
Canary Deployment (full)	Canary initiation to 100% traffic	4–24 hours	2–8 hours (AI-monitored auto-increment)
CVE Response (critical)	CVE detection to all consumers patched	7–90 days	< 24 hours (critical); < 72 hours (high)
Formulation Change (CPG/Pharma)	CCR to updated specification published	4–12 weeks	< 2 weeks (regulatory); < 3 days (internal)
Recall Scope Identification (CPG/Food)	Defect alert to affected batches listed	2–14 days	< 5 minutes
Financial Model Recalibration	Drift detection to re-calibrated model	2–8 weeks	< 5 business days (governed pipeline)
Tax Rule Update (legislative)	Legislation enacted to rule set deployed	6–10 weeks	< 2 weeks (effective-date scheduled release)
Farm-to-Consumer Traceability Query	Query to complete provenance report	2–7 days	< 10 minutes
Agricultural Seasonal Release	Season preparation to deployment	4–8 weeks	< 1 week (governed seasonal pipeline)

Knowledge Retrieval (Services)	Query to relevant methodology returned	30 minutes – 4 hours	< 2 minutes (AI semantic search)
Research Experiment Reproducibility	Attempt to reproduce vs. confirm	Often impossible	100% reproducible from stored artefact
Compliance Evidence Generation (SOC2)	Audit notice to evidence package ready	4–12 weeks	< 4 hours (automated generation)
Regulatory Examination Response	Examiner request to evidence produced	1–3 weeks	< 2 hours (direct query from platform)
Cross-Domain Security Incident	Detection to all sessions revoked	15–60 minutes	< 90 seconds (automated across all domains)
New Organisation Onboarding (multi-domain)	Contract signed to first SDE operational	4–12 weeks	< 1 business day

End of Document — Qala Universal Solution Management OS | Workflows & Use Cases v2.0